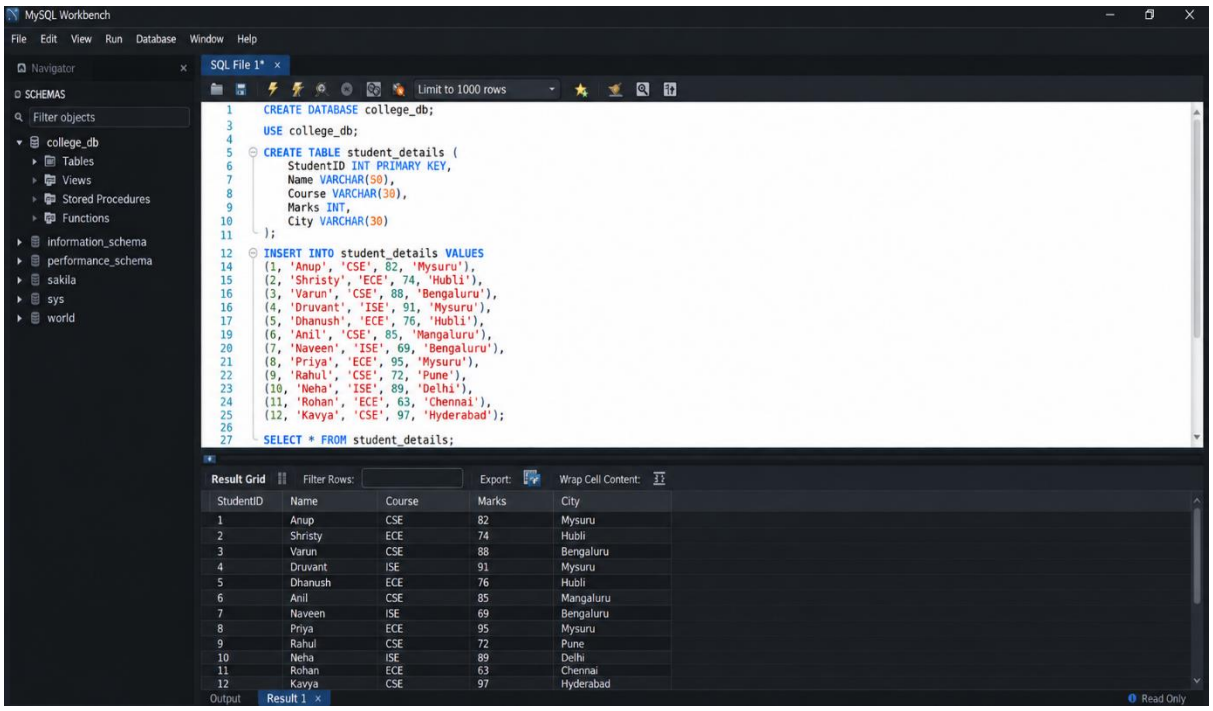
# AWS CLOUD & DATA ENGINEERING

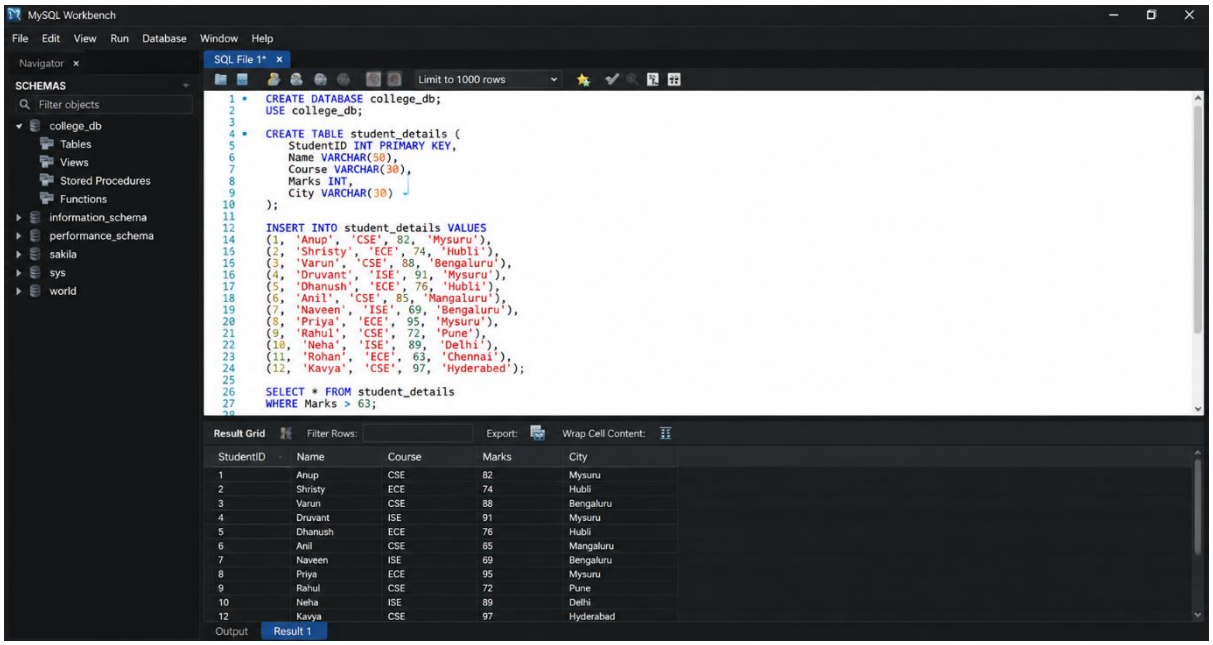
**NAME-ANUPKUMAR SHETTY**

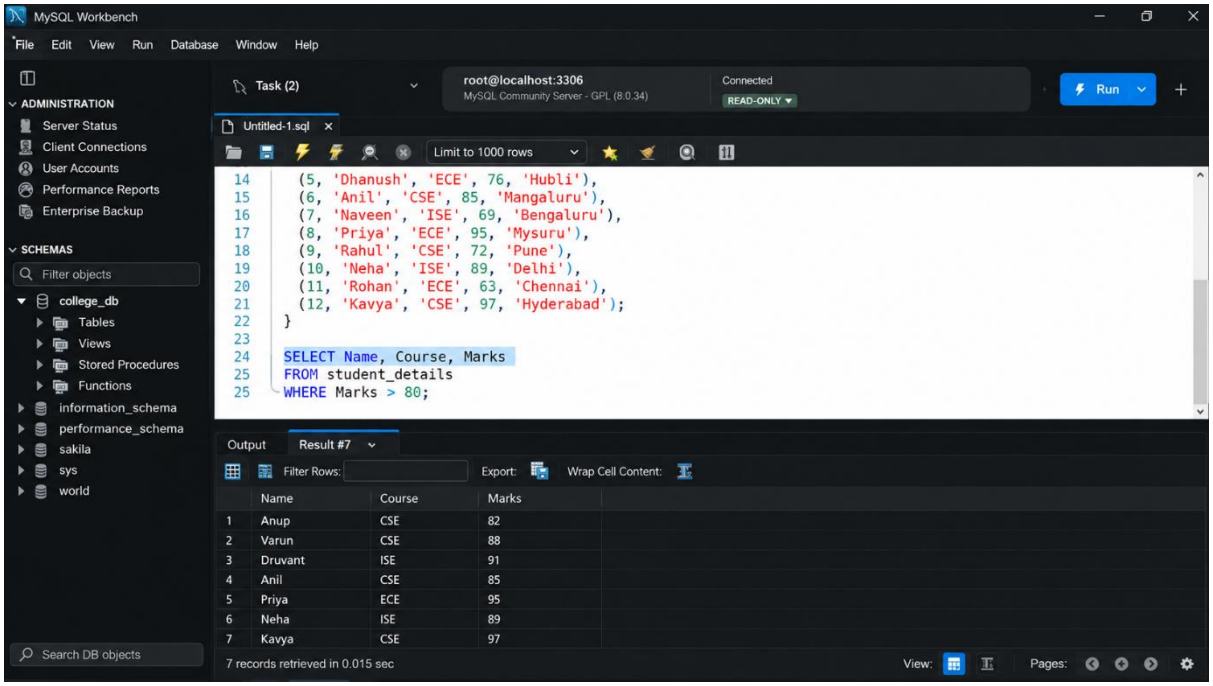
**USN-23BTRIS005**

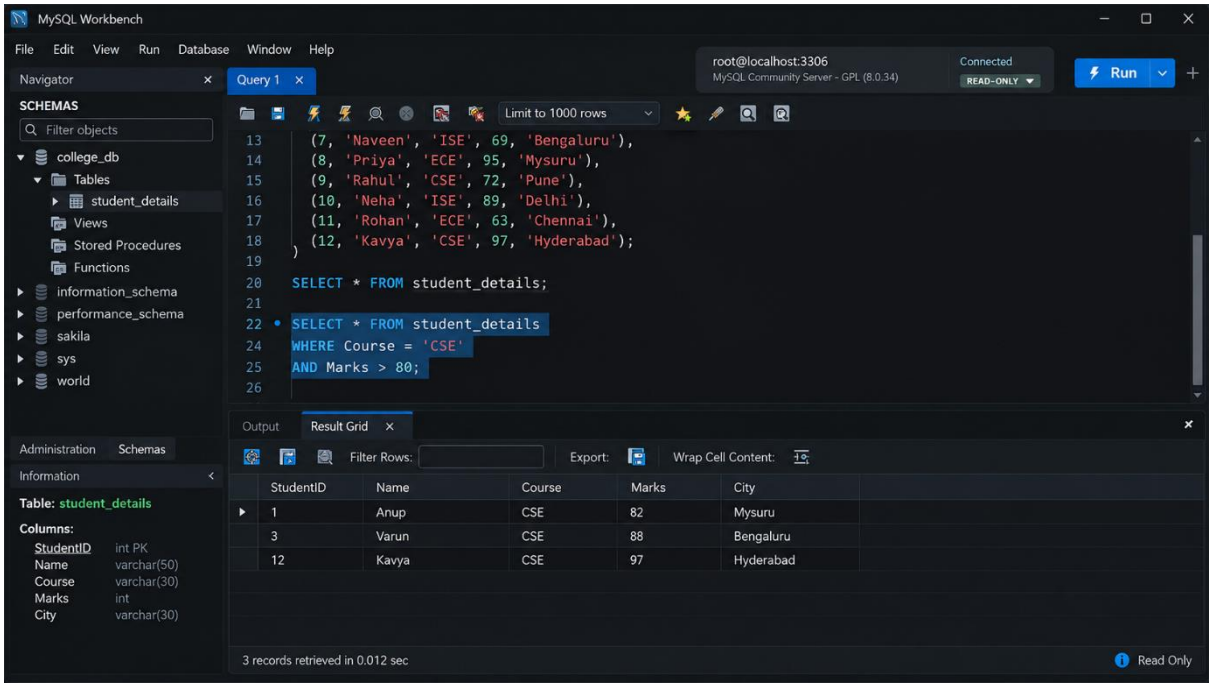
**COLLEGE-JAIN GLOBAL COLLEGE**

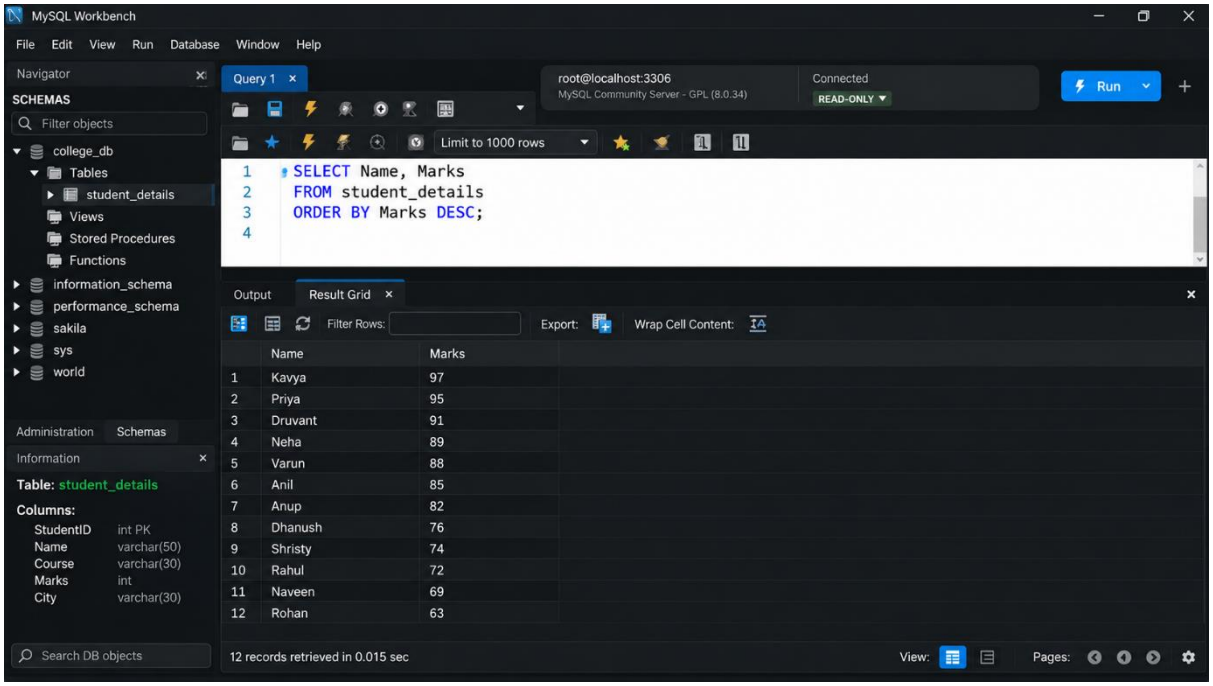
# SQL Practice: -

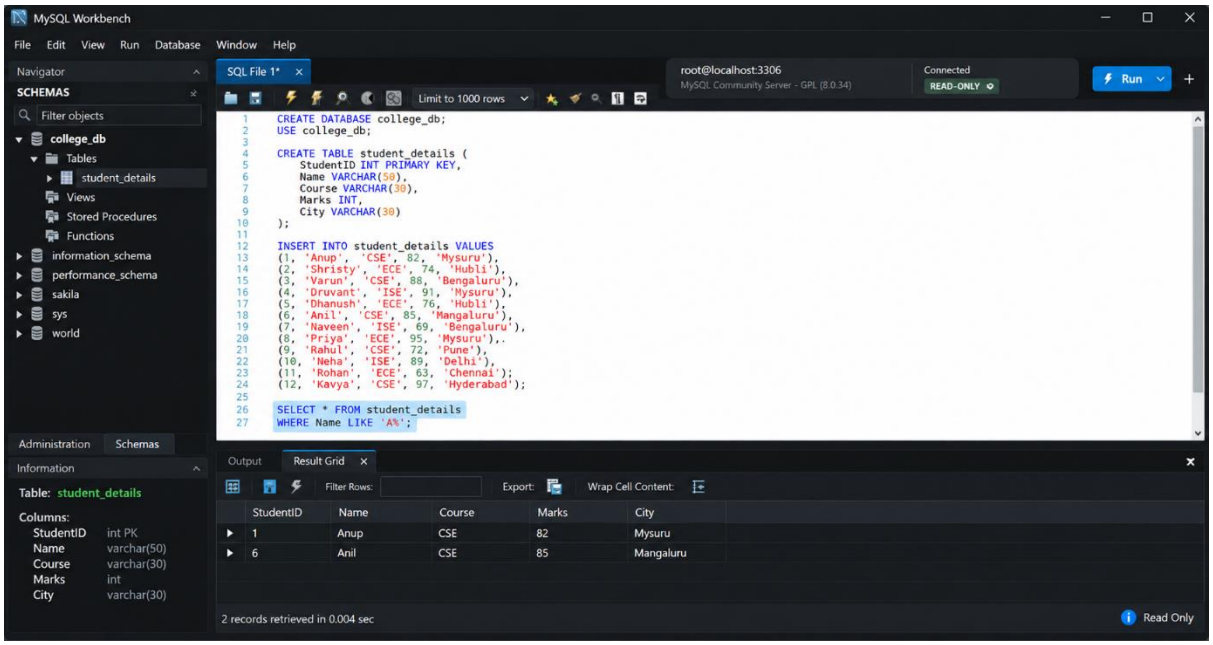


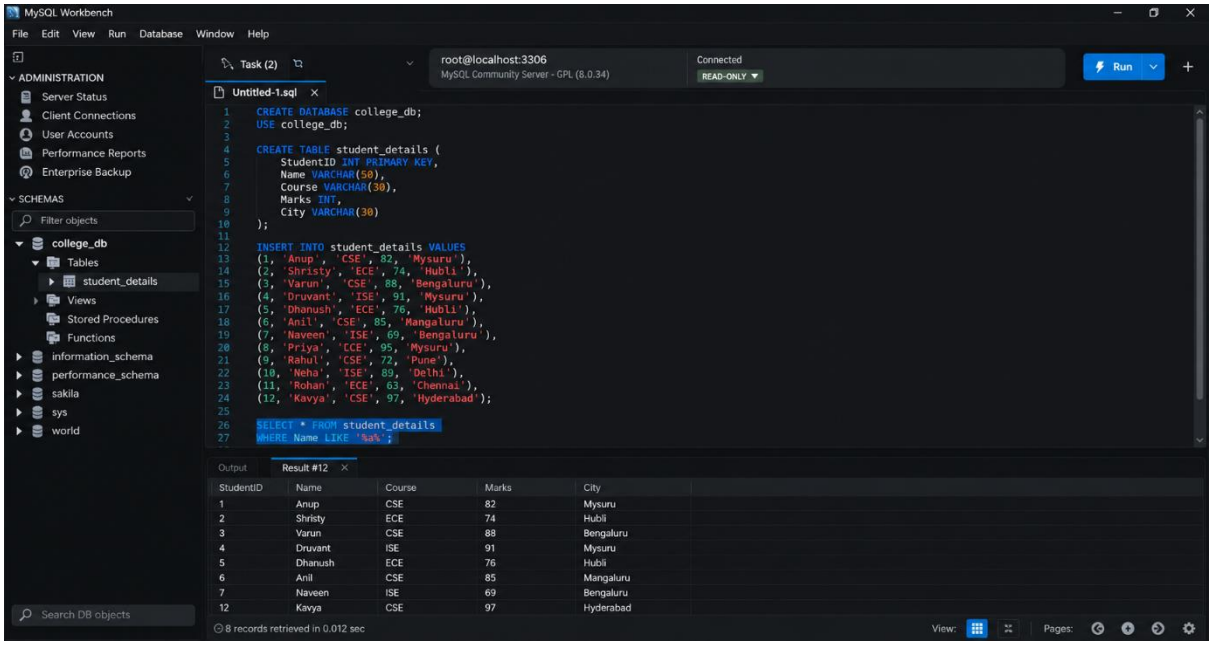



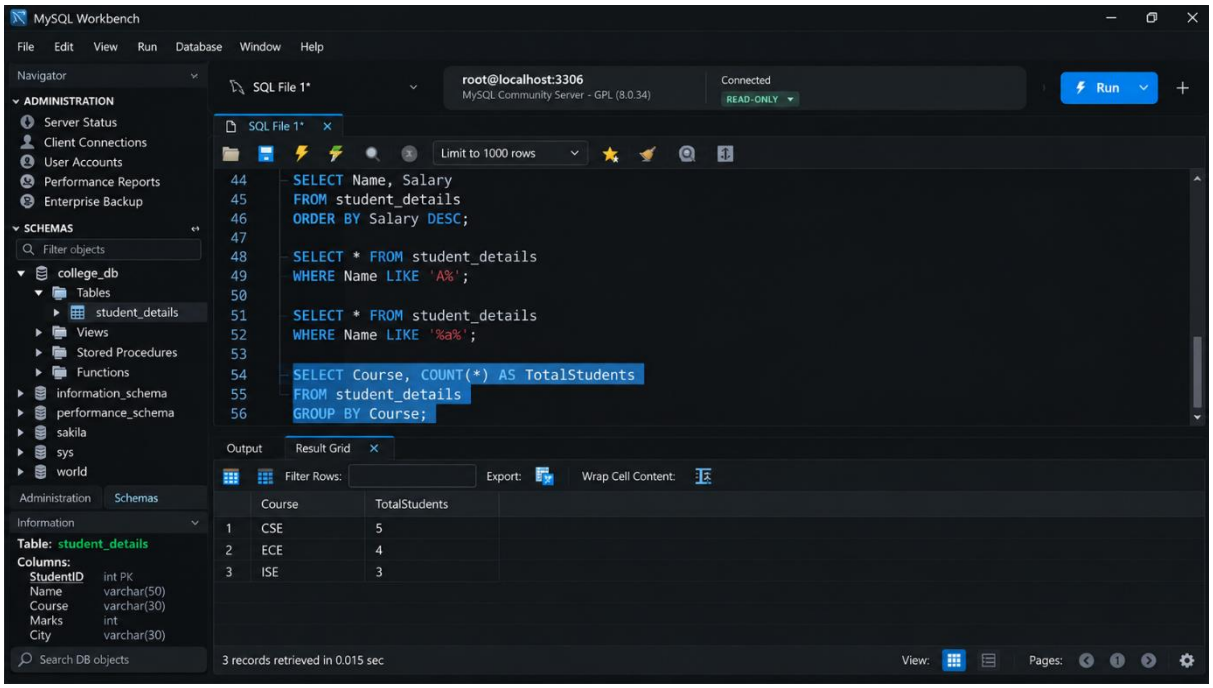

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Search DB objects
Task (2)
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Untitled-1.sql
Limit to 1000 rows
(5, 'Dhanush', 'ECE', 76, 'Hubli'),
(6, 'Anil', 'CSE', 85, 'Mangaluru'),
(7, 'Naveen', 'ISE', 69, 'Bengaluru'),
(8, 'Priya', 'ECE', 95, 'Mysuru'),
(9, 'Rahul', 'CSE', 72, 'Pune'),
(10, 'Neha', 'ISE', 89, 'Delhi'),
(11, 'Rohan', 'ECE', 63, 'Chennai'),
(12, 'Kavya', 'CSE', 97, 'Hyderabad');
}
SELECT Name, Course, Marks
FROM student_details
WHERE Marks > 80;
Output
Result #7
Filter Rows:
Export:
Wrap Cell Content:
Name Course Marks
1 Anup CSE 82
2 Varun CSE 88
3 Druvant ISE 91
4 Anil CSE 85
5 Priya ECE 95
6 Neha ISE 89
7 Kavya CSE 97
7 records retrieved in 0.015 sec
View:
Pages:

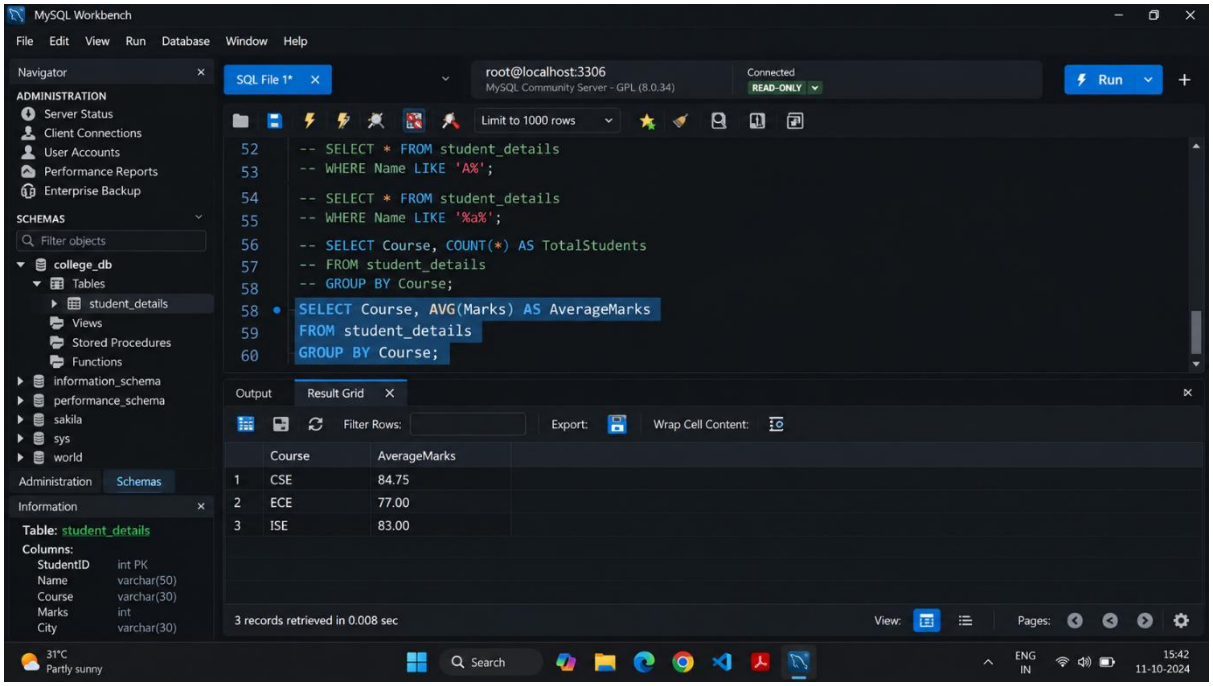

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Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
(7, 'Naveen', 'ISE', 69, 'Bengaluru'),
(8, 'Priya', 'ECE', 95, 'Mysuru'),
(9, 'Rahul', 'CSE', 72, 'Pune'),
(10, 'Neha', 'ISE', 89, 'Delhi'),
(11, 'Rohan', 'ECE', 63, 'Chennai'),
(12, 'Kavya', 'CSE', 97, 'Hyderabad');
)
SELECT * FROM student_details;
SELECT * FROM student_details
WHERE Course = 'CSE'
AND Marks > 80;
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
StudentID Name Course Marks City
1 Anup CSE 82 Mysuru
3 Varun CSE 88 Bengaluru
12 Kavya CSE 97 Hyderabad
3 records retrieved in 0.012 sec
Read Only

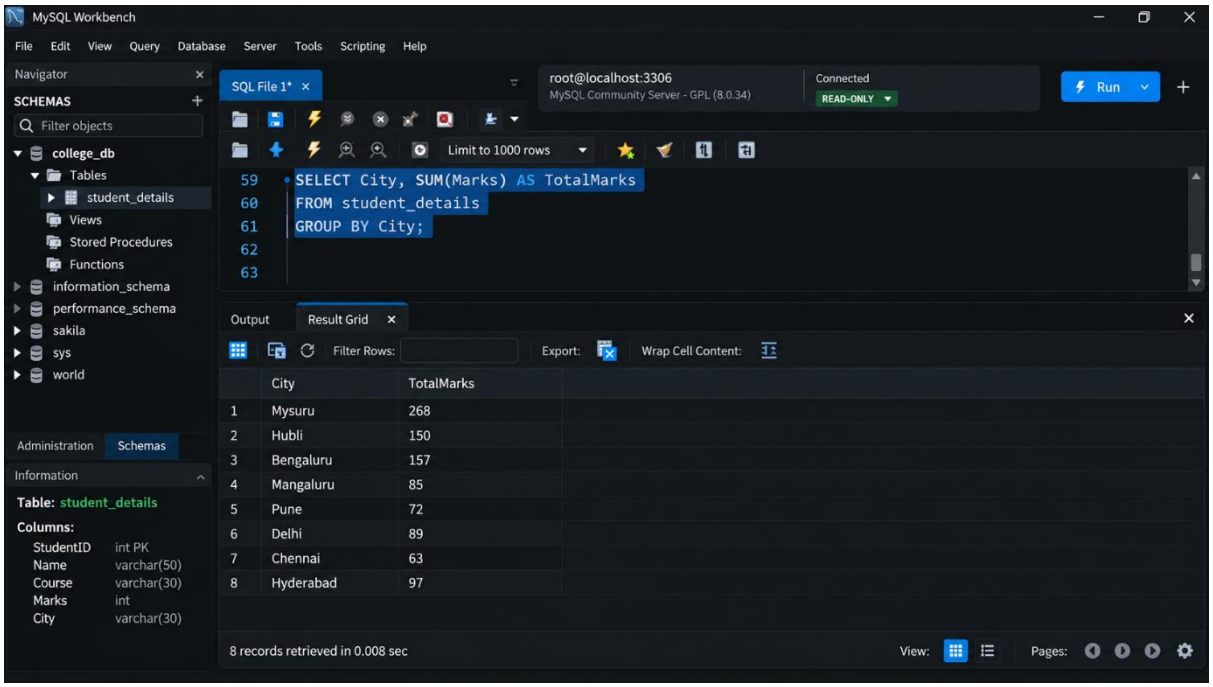

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Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
Search DB objects
Query 1
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
SELECT Name, Marks
FROM student_details
ORDER BY Marks DESC;
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Name Marks
1 Kavya 97
2 Priya 95
3 Druvant 91
4 Neha 89
5 Varun 88
6 Anil 85
7 Anup 82
8 Dhanush 76
9 Shristy 74
10 Rahul 72
11 Naveen 69
12 Rohan 63
12 records retrieved in 0.015 sec
View:
Pages:

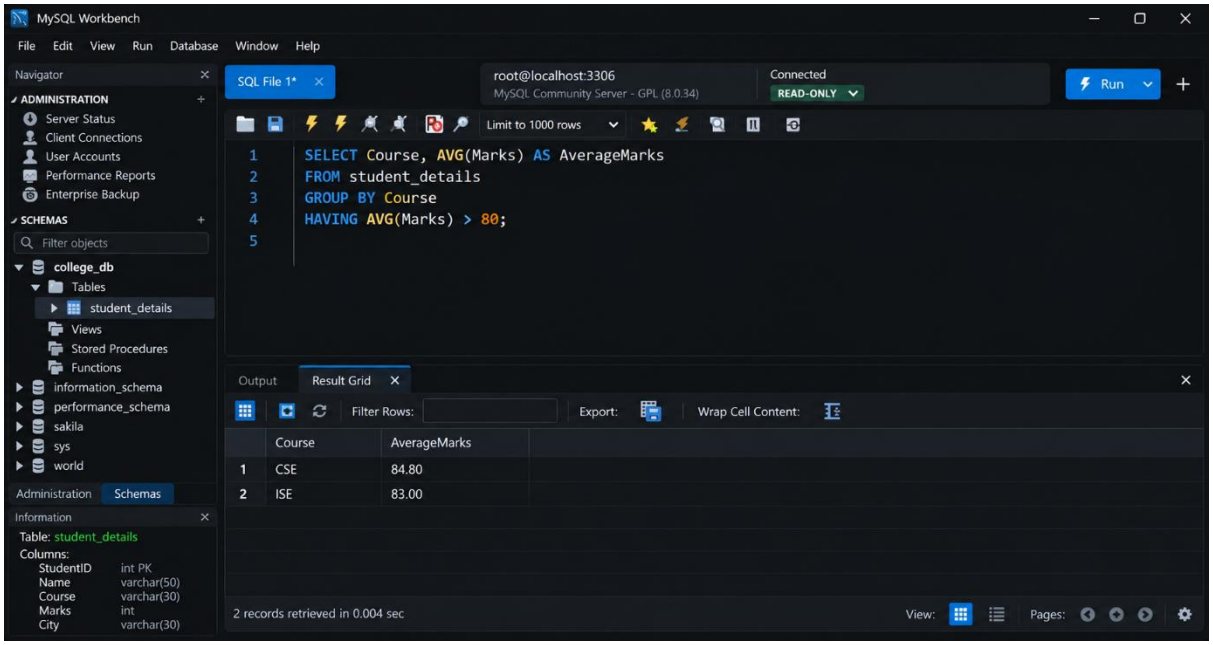

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Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
SQL File 1*
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
CREATE DATABASE college_db;
USE college_db;
CREATE TABLE student_details (
StudentID INT PRIMARY KEY,
Name VARCHAR(50),
Course VARCHAR(30),
Marks INT,
City VARCHAR(30)
);
INSERT INTO student_details VALUES
(1, 'Anup', 'CSE', 82, 'Mysuru'),
(2, 'Shristy', 'ECE', 74, 'Hubli'),
(3, 'Varun', 'CSE', 88, 'Bengaluru'),
(4, 'Druvant', 'ISE', 91, 'Mysuru'),
(5, 'Dhanush', 'ECE', 76, 'Hubli'),
(6, 'Anil', 'CSE', 85, 'Mangaluru'),
(7, 'Naveen', 'ISE', 69, 'Bengaluru'),
(8, 'Priya', 'ECE', 95, 'Mysuru'),.
(9, 'Rahul', 'CSE', 72, 'Pune'),
(10, 'Neha', 'ISE', 89, 'Delhi'),
(11, 'Rohan', 'ECE', 63, 'Chennai');
(12, 'Kavya', 'CSE', 97, 'Hyderabad');
SELECT * FROM student_details
WHERE Name LIKE 'A%';
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
StudentID Name Course Marks City
1 Anup CSE 82 Mysuru
6 Anil CSE 85 Mangaluru
2 records retrieved in 0.004 sec
Read Only

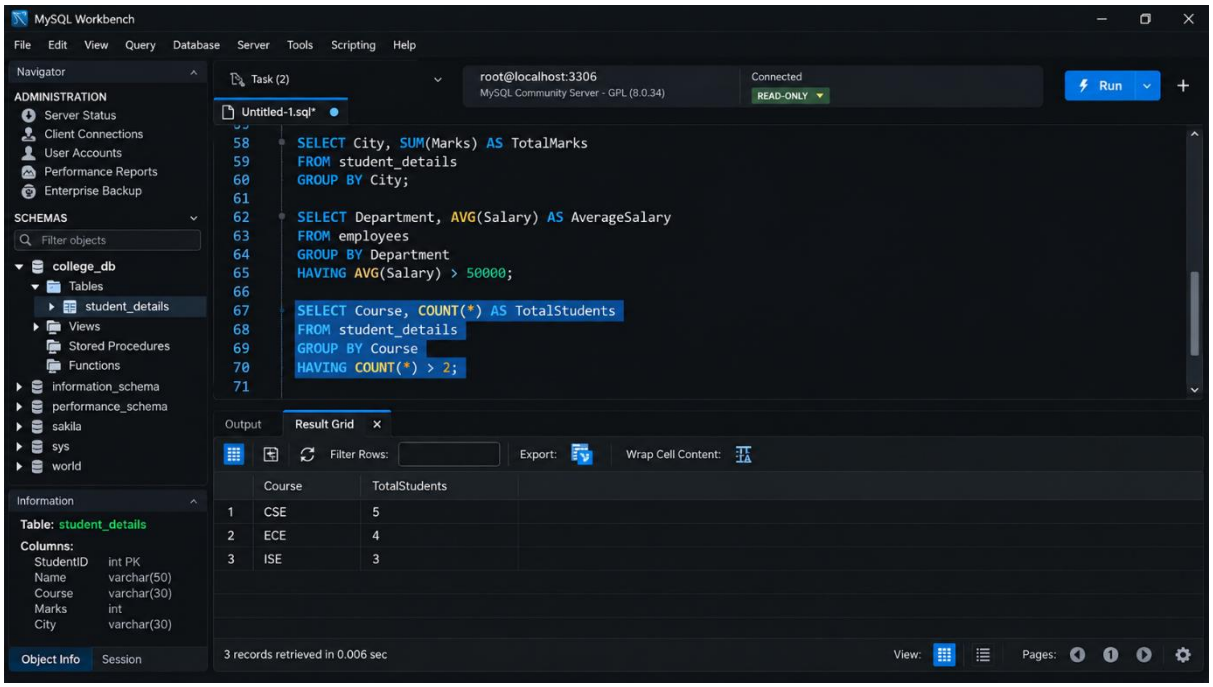

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Task (2)
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Untitled-1.sql
CREATE DATABASE college_db;
USE college_db;
CREATE TABLE student_details (
StudentID INT PRIMARY KEY,
Name VARCHAR(50),
Course VARCHAR(30),
Marks INT,
City VARCHAR(30)
);
INSERT INTO student_details VALUES
(1, 'Anup', 'CSE', 82, 'Mysuru'),
(2, 'Shristy', 'ECE', 74, 'Hubli'),
(3, 'Varun', 'CSE', 88, 'Bengaluru'),
(4, 'Druvant', 'ISE', 91, 'Mysuru'),
(5, 'Dhanush', 'ECE', 76, 'Hubli'),
(6, 'Anil', 'CSE', 85, 'Mangaluru'),
(7, 'Naveen', 'ISE', 69, 'Bengaluru'),
(8, 'Priya', 'CCE', 95, 'Mysuru'),
(9, 'Rahul', 'CSE', 72, 'Pune'),
(10, 'Neha', 'ISE', 89, 'Delhi'),
(11, 'Rohan', 'ECE', 63, 'Chennai'),
(12, 'Kavya', 'CSE', 97, 'Hyderabad');
SELECT * FROM student_details
WHERE Name LIKE '%a%';
Output
Result #12
StudentID Name Course Marks City
1 Anup CSE 82 Mysuru
2 Shristy ECE 74 Hubli
3 Varun CSE 88 Bengaluru
4 Druvant ISE 91 Mysuru
5 Dhanush ECE 76 Hubli
6 Anil CSE 85 Mangaluru
7 Naveen ISE 69 Bengaluru
12 Kavya CSE 97 Hyderabad
8 records retrieved in 0.012 sec
View:
Pages:

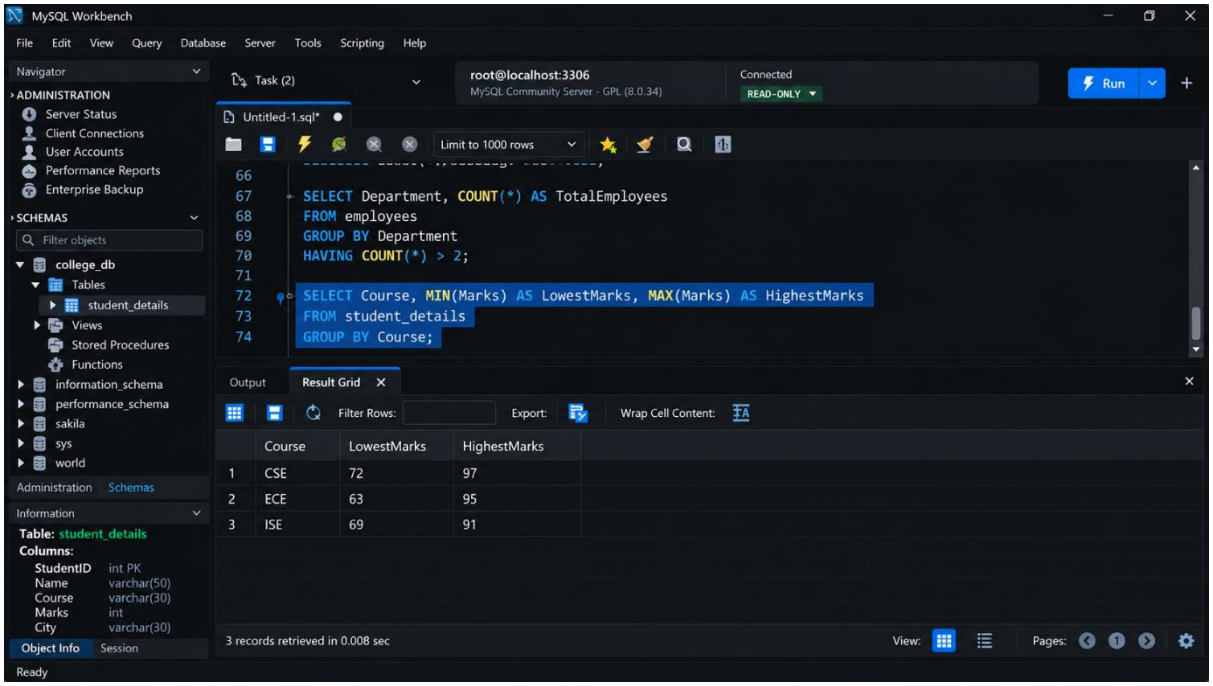

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Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
Search DB objects
SQL File 1*
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
SELECT Name, Salary
FROM student_details
ORDER BY Salary DESC;
SELECT * FROM student_details
WHERE Name LIKE 'A%';
SELECT * FROM student_details
WHERE Name LIKE '%a%';
SELECT Course, COUNT(*) AS TotalStudents
FROM student_details
GROUP BY Course;
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Course TotalStudents
1 CSE 5
2 ECE 4
3 ISE 3
3 records retrieved in 0.015 sec
View:
Pages:

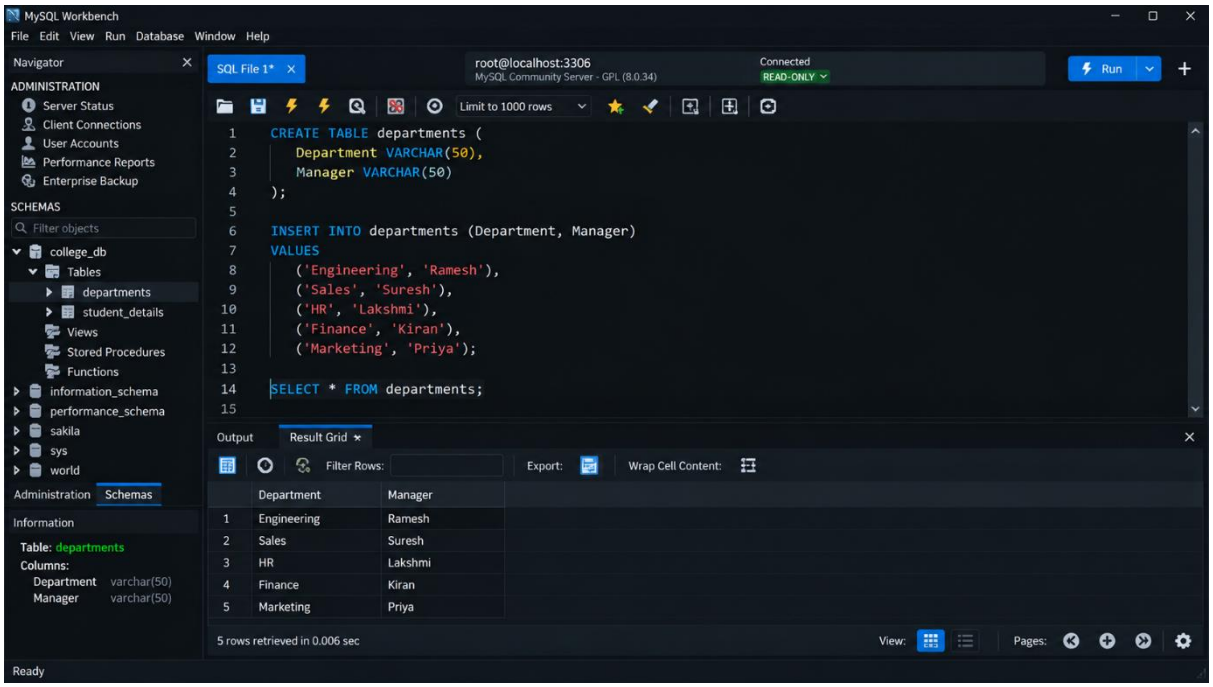

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Information
Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
SQL File 1*
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
52 -- SELECT * FROM student_details
53 -- WHERE Name LIKE 'A%';
54 -- SELECT * FROM student_details
55 -- WHERE Name LIKE '%a%';
56 -- SELECT Course, COUNT(*) AS TotalStudents
57 -- FROM student_details
58 -- GROUP BY Course;
58 SELECT Course, AVG(Marks) AS AverageMarks
59 FROM student_details
60 GROUP BY Course;
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Course AverageMarks
1 CSE 84.75
2 ECE 77.00
3 ISE 83.00
3 records retrieved in 0.008 sec
View:
Pages:
31°C
Partly sunny
Search
ENG
IN
15:42
11-10-2024

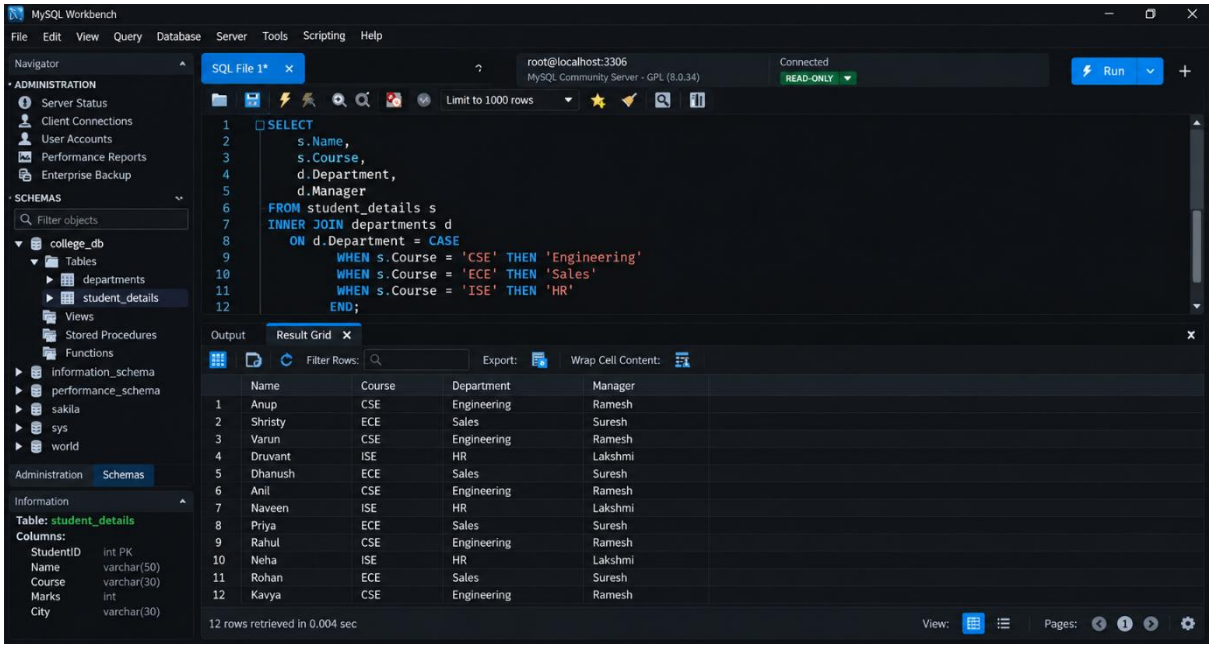

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Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
SQL File 1*
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
59 SELECT City, SUM(Marks) AS TotalMarks
60 FROM student_details
61 GROUP BY City;
62
63
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
City TotalMarks
1 Mysuru 268
2 Hubli 150
3 Bengaluru 157
4 Mangaluru 85
5 Pune 72
6 Delhi 89
7 Chennai 63
8 Hyderabad 97
8 records retrieved in 0.008 sec
View:
Pages:

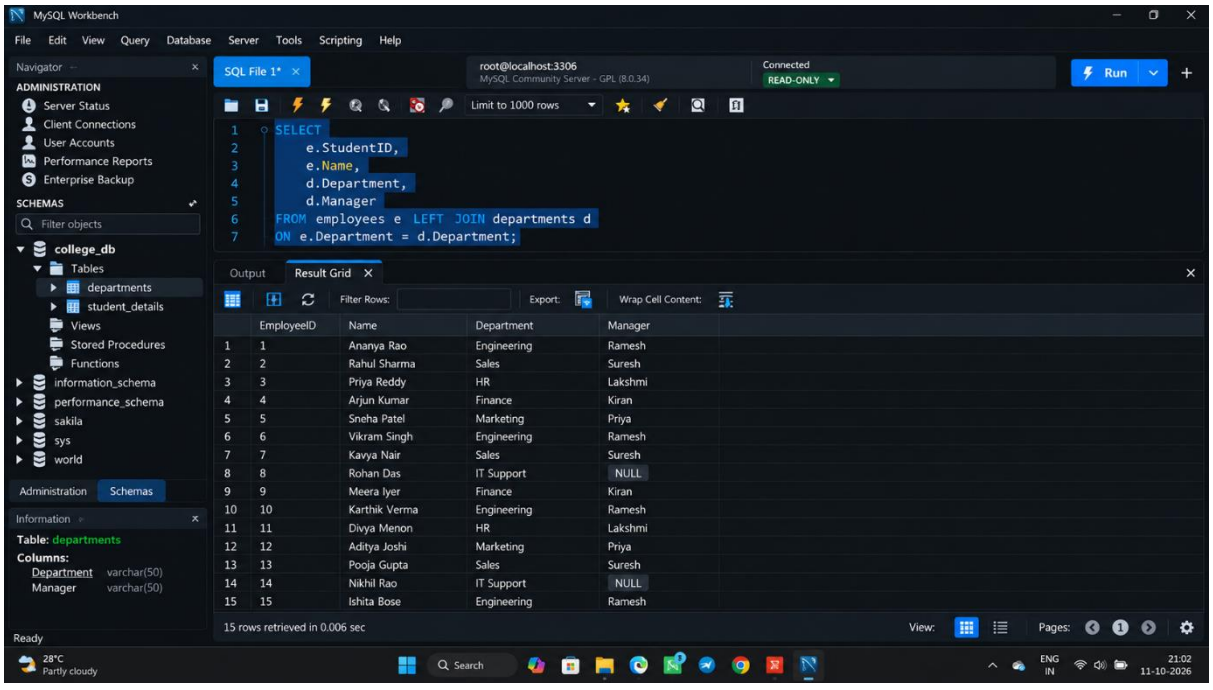

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Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
SQL File 1*
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
1 SELECT Course, AVG(Marks) AS AverageMarks
2 FROM student_details
3 GROUP BY Course
4 HAVING AVG(Marks) > 80;
5
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Course
AverageMarks
1 CSE 84.80
2 ISE 83.00
2 records retrieved in 0.004 sec
View:
Pages:

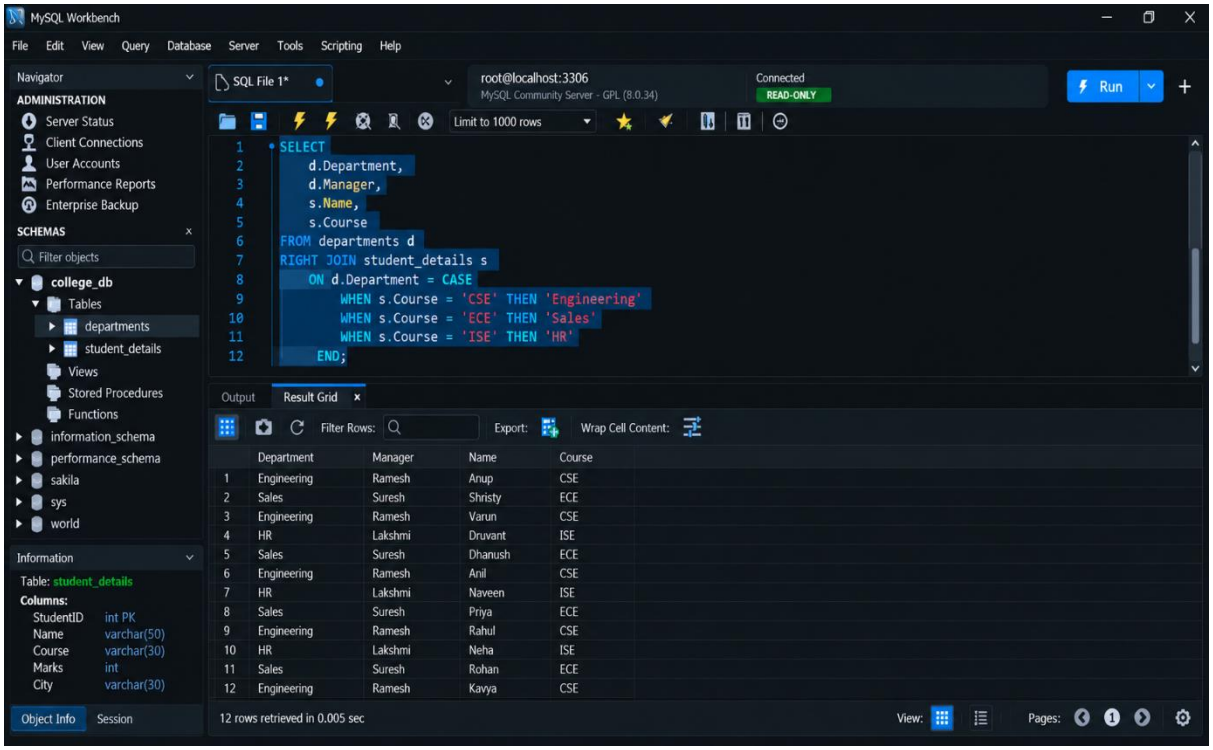

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Information
Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
Object Info Session
Task (2)
Untitled-1.sql*
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
58 SELECT City, SUM(Marks) AS TotalMarks
59 FROM student_details
60 GROUP BY City;
61
62 SELECT Department, AVG(Salary) AS AverageSalary
63 FROM employees
64 GROUP BY Department
65 HAVING AVG(Salary) > 50000;
66
67 SELECT Course, COUNT(*) AS TotalStudents
68 FROM student_details
69 GROUP BY Course
70 HAVING COUNT(*) > 2;
71
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Course
TotalStudents
1 CSE 5
2 ECE 4
3 ISE 3
3 records retrieved in 0.006 sec
View:
Pages:


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Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
Object Info Session
Task (2)
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Untitled-1.sql*
Limit to 1000 rows
66
67 SELECT Department, COUNT(*) AS TotalEmployees
68 FROM employees
69 GROUP BY Department
70 HAVING COUNT(*) > 2;
71
72 SELECT Course, MIN(Marks) AS LowestMarks, MAX(Marks) AS HighestMarks
73 FROM student_details
74 GROUP BY Course;
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Course LowestMarks HighestMarks
1 CSE 72 97
2 ECE 63 95
3 ISE 69 91
3 records retrieved in 0.008 sec
View:
Pages:
Ready

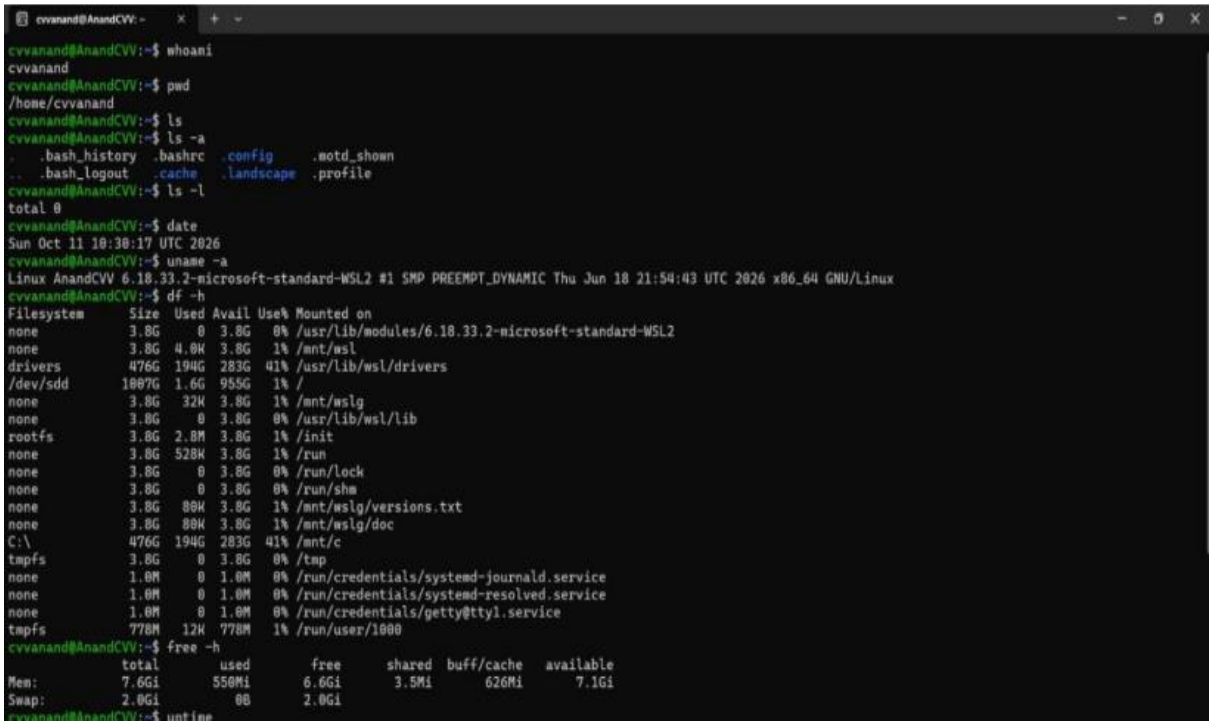

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Table: departments
Columns:
Department varchar(50)
Manager varchar(50)
SQL File 1*
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
1 CREATE TABLE departments (
2 Department VARCHAR(50),
3 Manager VARCHAR(50)
4 );
5
6 INSERT INTO departments (Department, Manager)
7 VALUES
8 ('Engineering', 'Ramesh'),
9 ('Sales', 'Suresh'),
10 ('HR', 'Lakshmi'),
11 ('Finance', 'Kiran'),
12 ('Marketing', 'Priya');
13
14 SELECT * FROM departments;
15
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Department Manager
1 Engineering Ramesh
2 Sales Suresh
3 HR Lakshmi
4 Finance Kiran
5 Marketing Priya
5 rows retrieved in 0.006 sec
View:
Pages:
Ready


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Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
SQL File 1*
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
SELECT
s.Name,
s.Course,
d.Department,
d.Manager
FROM student_details s
INNER JOIN departments d
ON d.Department = CASE
WHEN s.Course = 'CSE' THEN 'Engineering'
WHEN s.Course = 'ECE' THEN 'Sales'
WHEN s.Course = 'ISE' THEN 'HR'
END;
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Name Course Department Manager
1 Anup CSE Engineering Ramesh
2 Shristy ECE Sales Suresh
3 Varun CSE Engineering Ramesh
4 Druvant ISE HR Lakshmi
5 Dhanush ECE Sales Suresh
6 Anil CSE Engineering Ramesh
7 Naveen ISE HR Lakshmi
8 Priya ECE Sales Suresh
9 Rahul CSE Engineering Ramesh
10 Neha ISE HR Lakshmi
11 Rohan ECE Sales Suresh
12 Kavya CSE Engineering Ramesh
12 rows retrieved in 0.004 sec
View:
Pages:

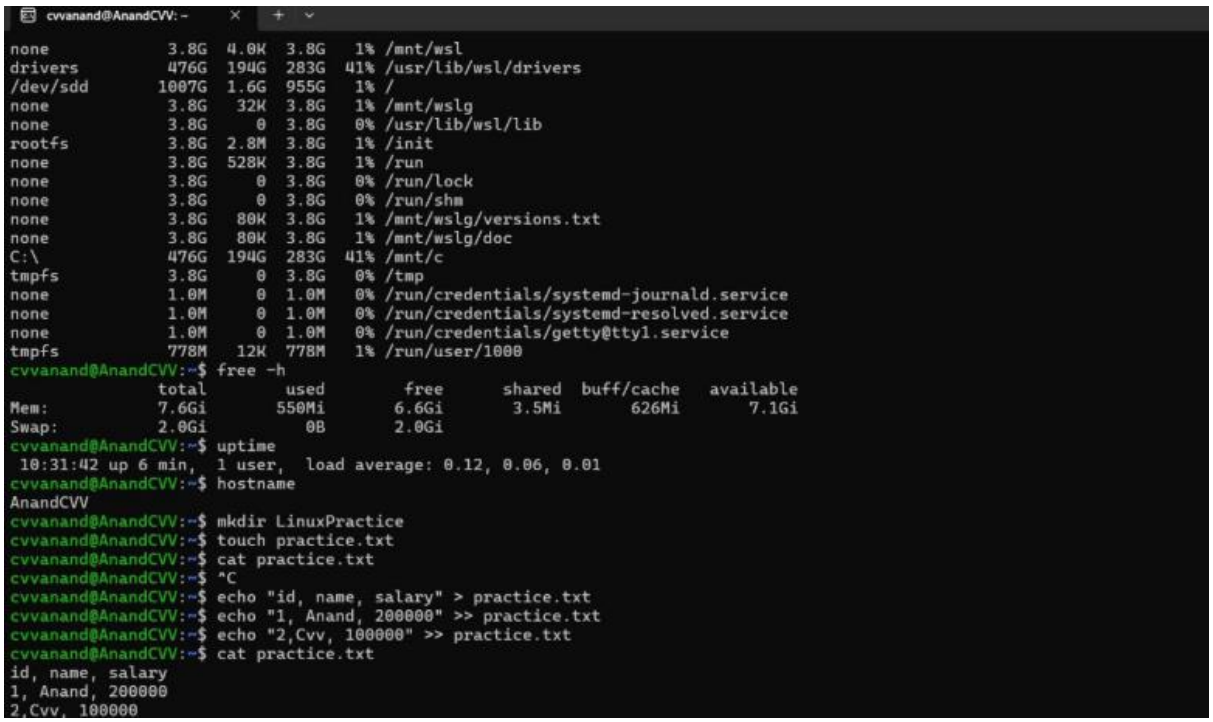

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Table: departments
Columns:
Department varchar(50)
Manager varchar(50)
SQL File 1*
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
SELECT
e.StudentID,
e.Name,
d.Department,
d.Manager
FROM employees e LEFT JOIN departments d
ON e.Department = d.Department;
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
EmployeeID Name Department Manager
1 1 Ananya Rao Engineering Ramesh
2 2 Rahul Sharma Sales Suresh
3 3 Priya Reddy HR Lakshmi
4 4 Arjun Kumar Finance Kiran
5 5 Sneha Patel Marketing Priya
6 6 Vikram Singh Engineering Ramesh
7 7 Kavya Nair Sales Suresh
8 8 Rohan Das IT Support NULL
9 9 Meera Iyer Finance Kiran
10 10 Karthik Verma Engineering Ramesh
11 11 Divya Menon HR Lakshmi
12 12 Aditya Joshi Marketing Priya
13 13 Pooja Gupta Sales Suresh
14 14 Nikhil Rao IT Support NULL
15 15 Ishita Bose Engineering Ramesh
15 rows retrieved in 0.006 sec
View:
Pages:
Ready
28°C
Partly cloudy
Search
ENG
IN
21:02
11-10-2026


MySQL Workbench
File Edit View Query Database Server Tools Scripting Help
Navigator
ADMINISTRATION
Server Status
Client Connections
User Accounts
Performance Reports
Enterprise Backup
SCHEMAS
Filter objects
college_db
Tables
departments
student_details
Views
Stored Procedures
Functions
information_schema
performance_schema
sakila
sys
world
Information
Table: student_details
Columns:
StudentID int PK
Name varchar(50)
Course varchar(30)
Marks int
City varchar(30)
Object Info
Session
SQL File 1*
root@localhost:3306
MySQL Community Server - GPL (8.0.34)
Connected
READ-ONLY
Run
Limit to 1000 rows
SELECT
d.Department,
d.Manager,
s.Name,
s.Course
FROM departments d
RIGHT JOIN student_details s
ON d.Department = CASE
WHEN s.Course = 'CSE' THEN 'Engineering'
WHEN s.Course = 'ECE' THEN 'Sales'
WHEN s.Course = 'ISE' THEN 'HR'
END;
Output
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Department Manager Name Course
1 Engineering Ramesh Anup CSE
2 Sales Suresh Shristy ECE
3 Engineering Ramesh Varun CSE
4 HR Lakshmi Druvant ISE
5 Sales Suresh Dhanush ECE
6 Engineering Ramesh Anil CSE
7 HR Lakshmi Naveen ISE
8 Sales Suresh Priya ECE
9 Engineering Ramesh Rahul CSE
10 HR Lakshmi Neha ISE
11 Sales Suresh Rohan ECE
12 Engineering Ramesh Kavya CSE
12 rows retrieved in 0.005 sec
View:
Pages:

# Linux Practice: -

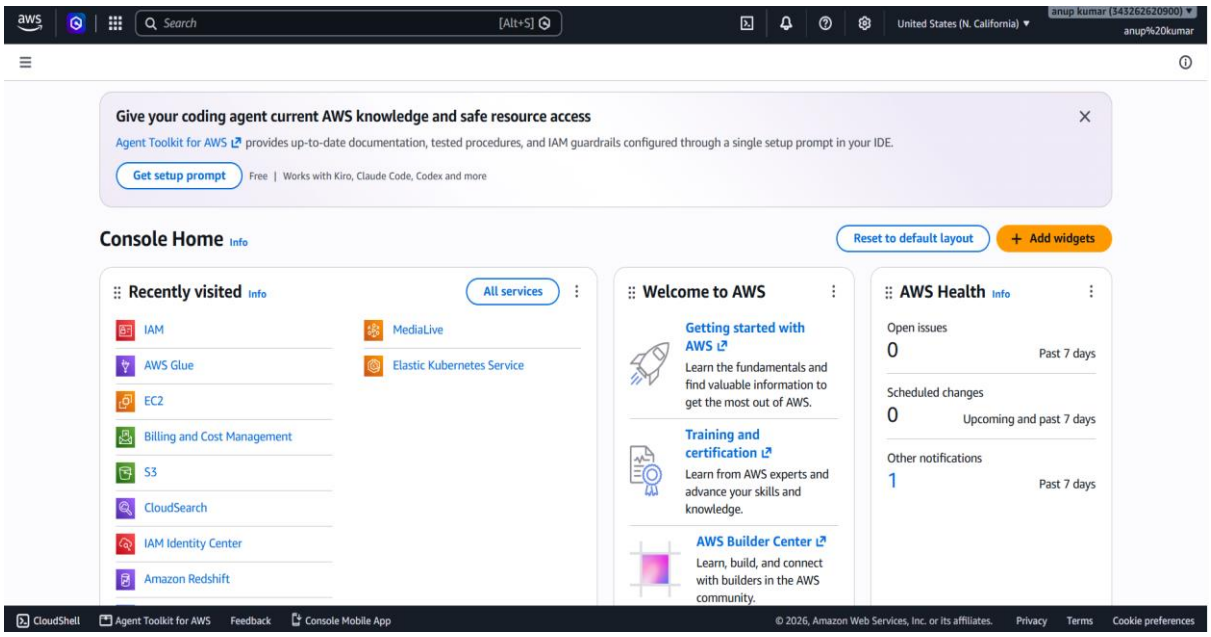


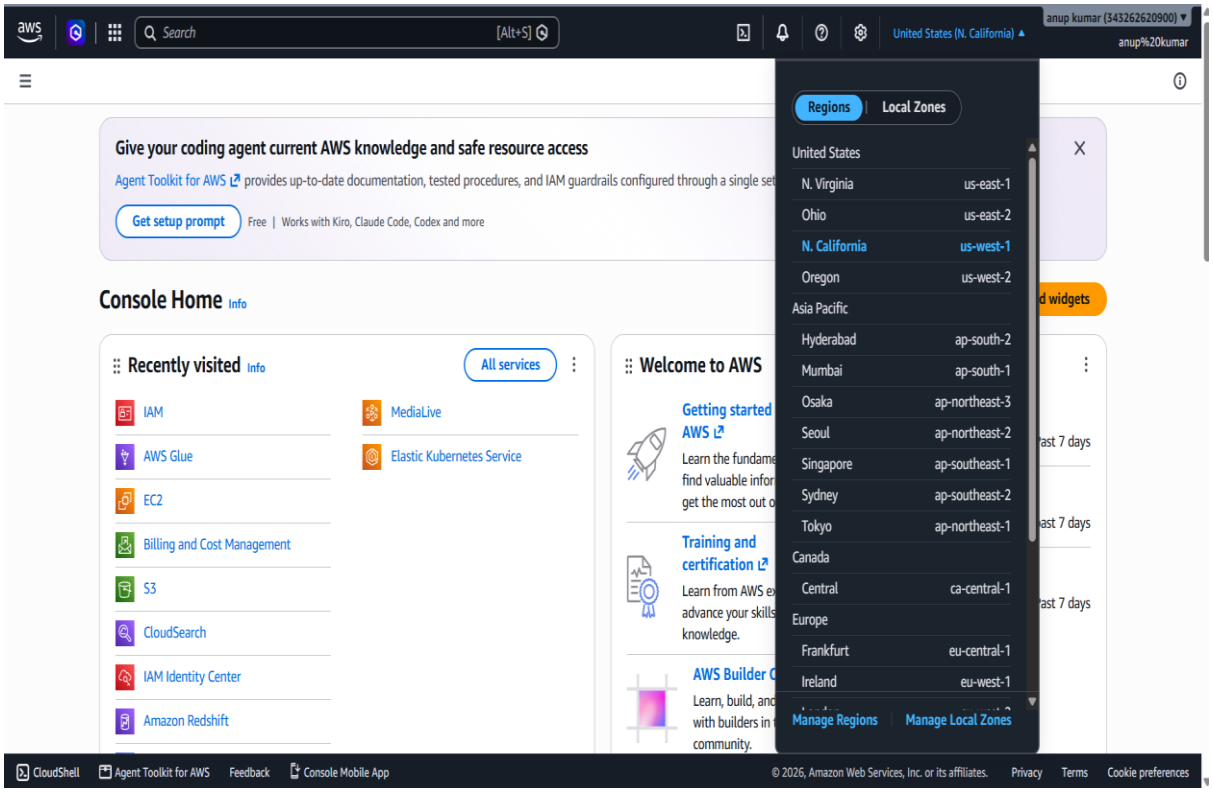
```
cvvanand@AnandCVV:~$ whoami
cvvanand
cvvanand@AnandCVV:~$ pwd
/home/cvvanand
cvvanand@AnandCVV:~$ ls
cvvanand@AnandCVV:~$ ls -a
.   .bash_history  .bashrc  .config     .motd_shown
..  .bash_logout   .cache   .landscape  .profile
cvvanand@AnandCVV:~$ ls -l
total 0
cvvanand@AnandCVV:~$ date
Sun Oct 11 10:30:17 UTC 2026
cvvanand@AnandCVV:~$ uname -a
Linux AnandCVV 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
cvvanand@AnandCVV:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.8G     0  3.8G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.8G  4.0K  3.8G   1% /mnt/wsl
drivers         476G  194G  283G  41% /usr/lib/wsl/drivers
/dev/sdd       1007G  1.6G  955G   1% /
none            3.8G   32K  3.8G   1% /mnt/wslg
none            3.8G     0  3.8G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M  3.8G   1% /init
none            3.8G  528K  3.8G   1% /run
none            3.8G     0  3.8G   0% /run/lock
none            3.8G     0  3.8G   0% /run/shm
none            3.8G   80K  3.8G   1% /mnt/wslg/versions.txt
none            3.8G   80K  3.8G   1% /mnt/wslg/doc
C:\             476G  194G  283G  41% /mnt/c
tmpfs           3.8G     0  3.8G   0% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           778M   12K  778M   1% /run/user/1000
cvvanand@AnandCVV:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       550Mi       6.6Gi       3.5Mi       626Mi       7.1Gi
Swap:          2.0Gi          0B       2.0Gi
cvvanand@AnandCVV:~$ uptime
```



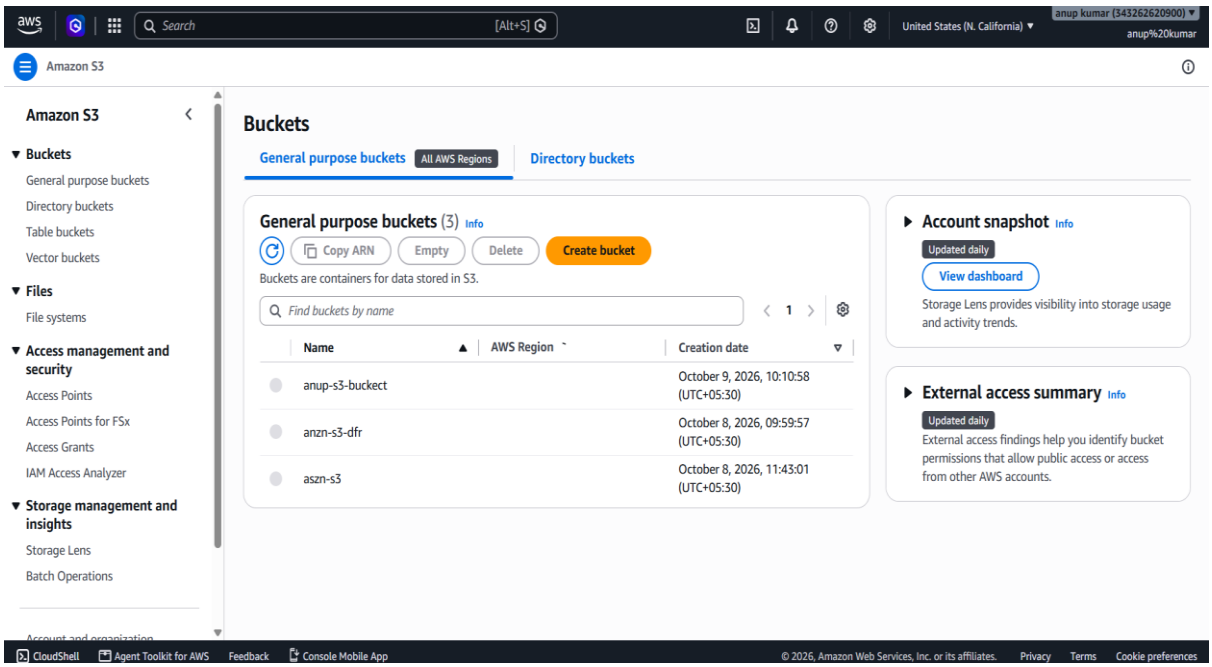
```
none            3.8G  4.0K  3.8G   1% /mnt/wsl
drivers         476G  194G  283G  41% /usr/lib/wsl/drivers
/dev/sdd       1007G  1.6G  955G   1% /
none            3.8G   32K  3.8G   1% /mnt/wslg
none            3.8G     0  3.8G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M  3.8G   1% /init
none            3.8G  528K  3.8G   1% /run
none            3.8G     0  3.8G   0% /run/lock
none            3.8G     0  3.8G   0% /run/shm
none            3.8G   80K  3.8G   1% /mnt/wslg/versions.txt
none            3.8G   80K  3.8G   1% /mnt/wslg/doc
C:\             476G  194G  283G  41% /mnt/c
tmpfs           3.8G     0  3.8G   0% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           778M   12K  778M   1% /run/user/1000
cvvanand@AnandCVV:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       550Mi       6.6Gi       3.5Mi       626Mi       7.1Gi
Swap:          2.0Gi          0B       2.0Gi
cvvanand@AnandCVV:~$ uptime
 10:31:42 up 6 min,  1 user,  load average: 0.12, 0.06, 0.01
cvvanand@AnandCVV:~$ hostname
AnandCVV
cvvanand@AnandCVV:~$ mkdir LinuxPractice
cvvanand@AnandCVV:~$ touch practice.txt
cvvanand@AnandCVV:~$ cat practice.txt
cvvanand@AnandCVV:~$ ^C
cvvanand@AnandCVV:~$ echo "id, name, salary" > practice.txt
cvvanand@AnandCVV:~$ echo "1, Anand, 200000" >> practice.txt
cvvanand@AnandCVV:~$ echo "2,Cvv, 100000" >> practice.txt
cvvanand@AnandCVV:~$ cat practice.txt
id, name, salary
1, Anand, 200000
2,Cvv, 100000
```

# AWS Account / Region: -

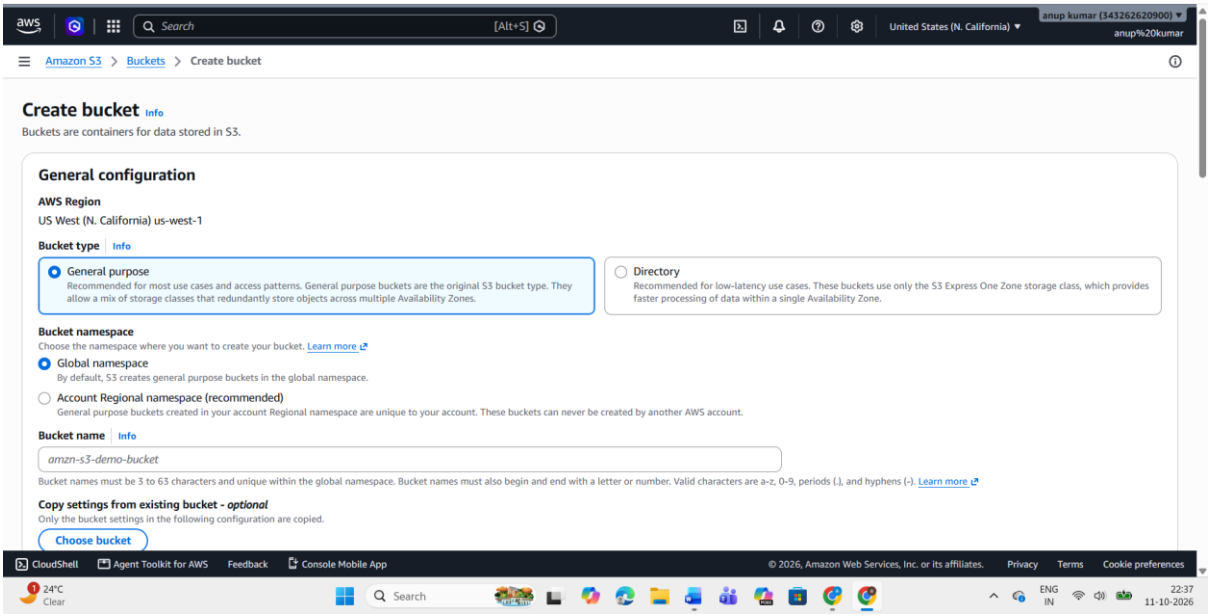


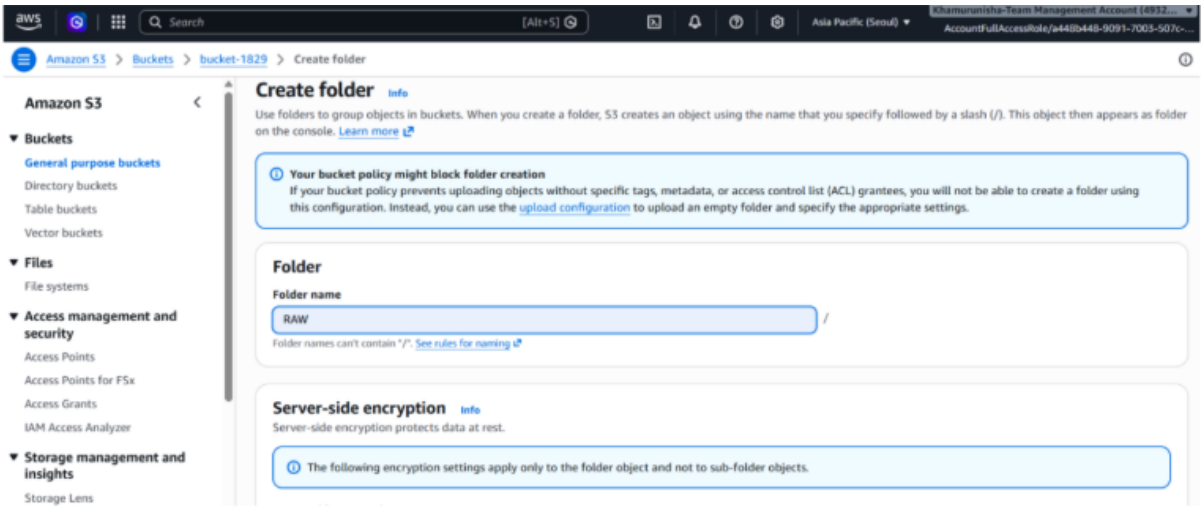
# Amazon S3



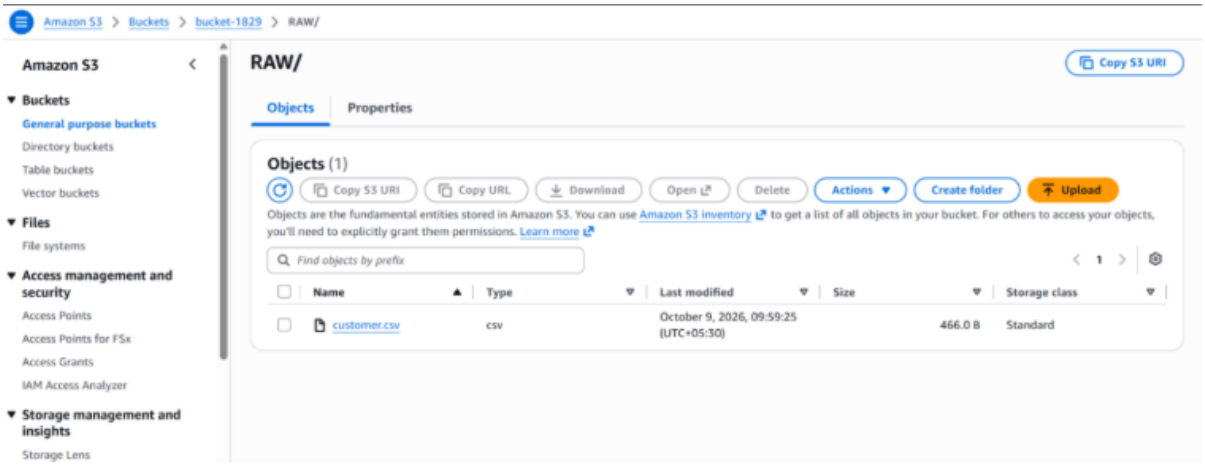

## 1. Create an S3 bucket



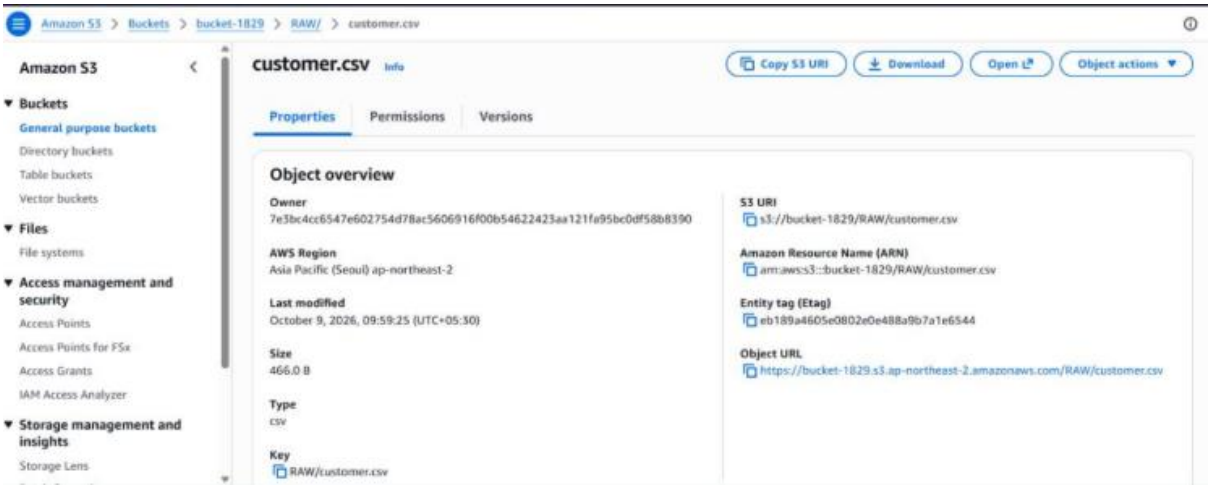
## 2. Create the raw/ folder



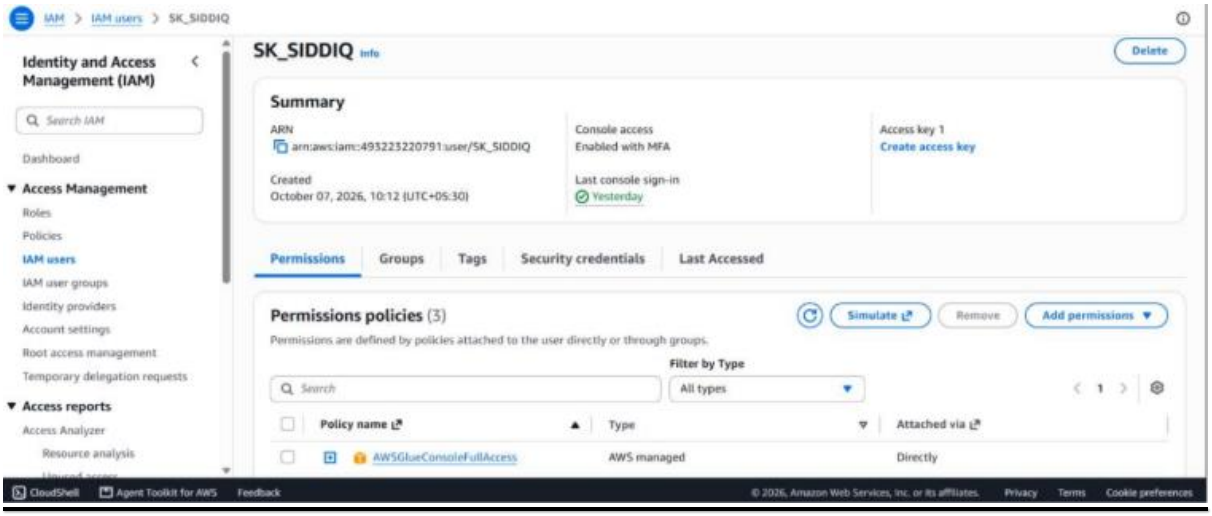

## 3. Upload the file into raw



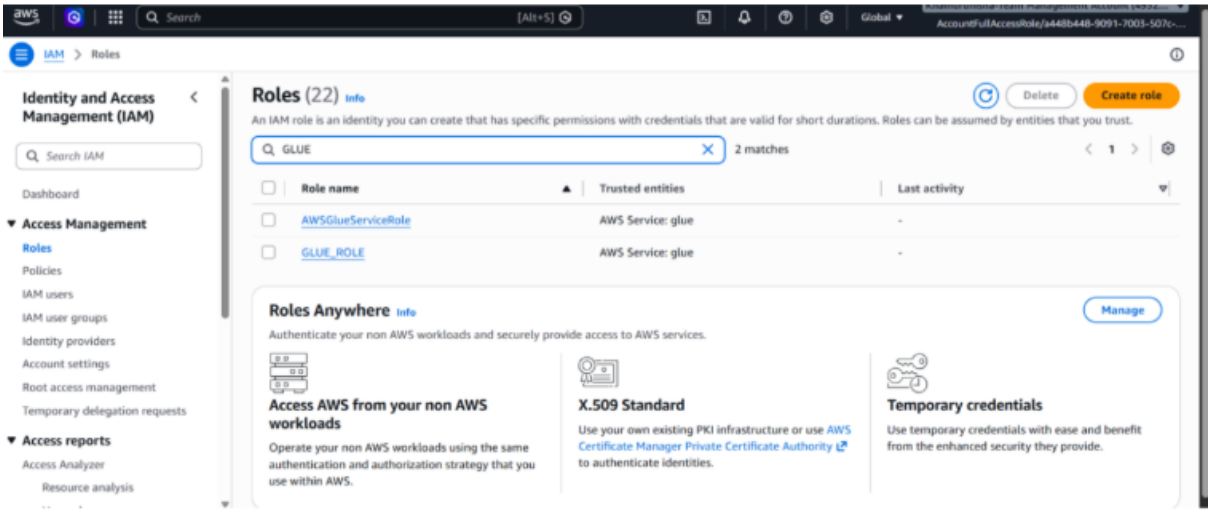

## 4. s3://bucket-1829/RAW/customer.csv



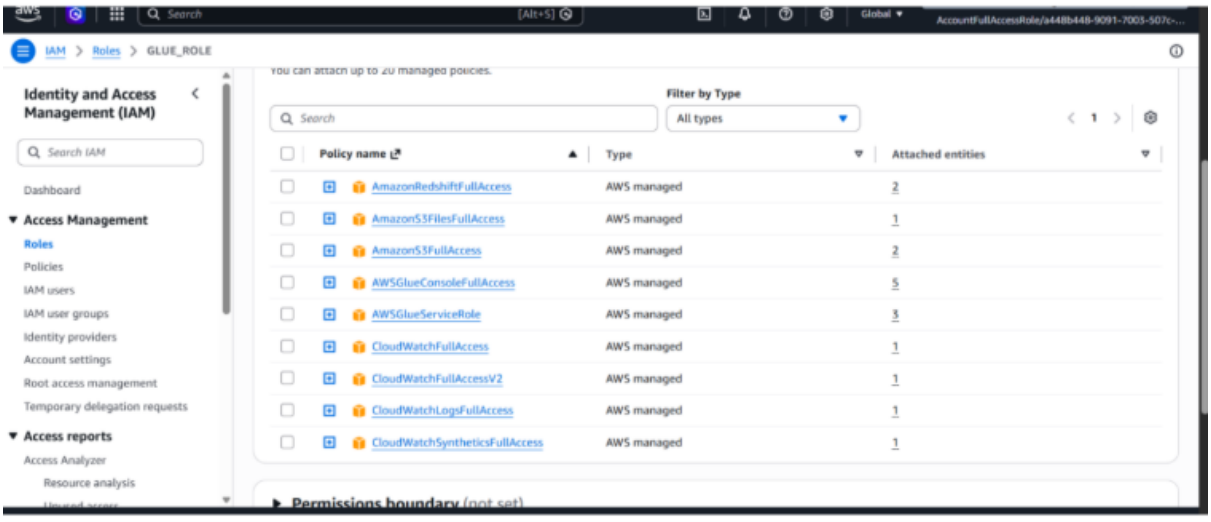
# IAM(IDENTITY-ACCESS-MANAGEMENT)




# GLUE-ROLE

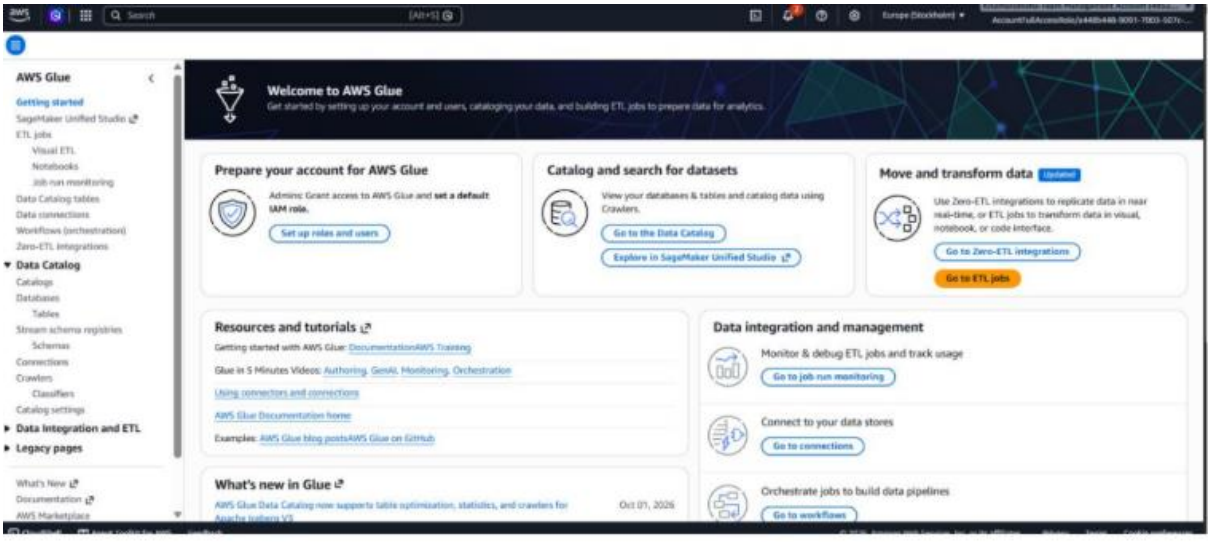


# GLUE-ROLE-policies

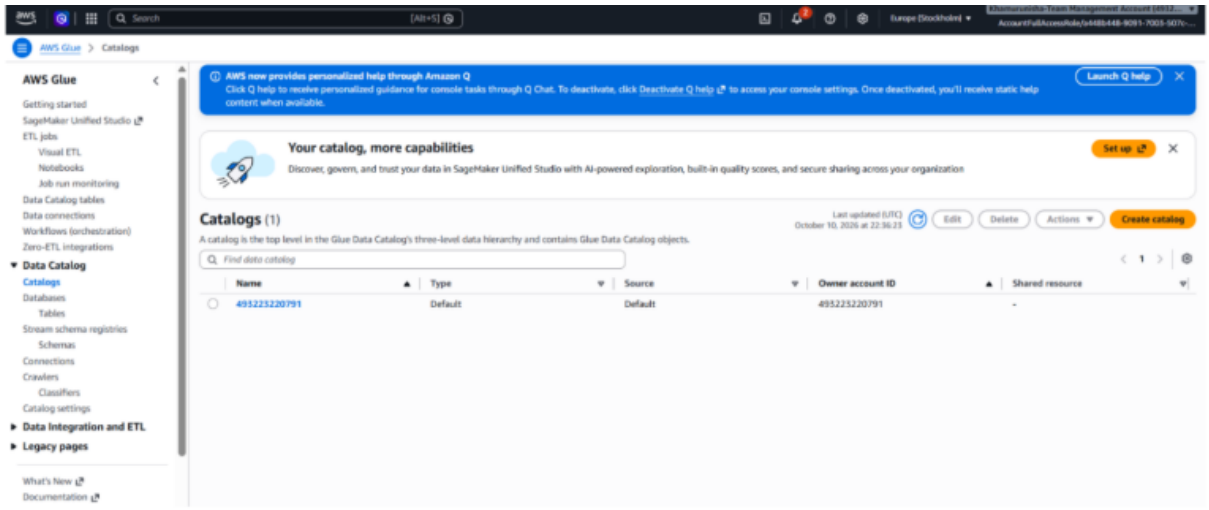



# AWS Glue Database:

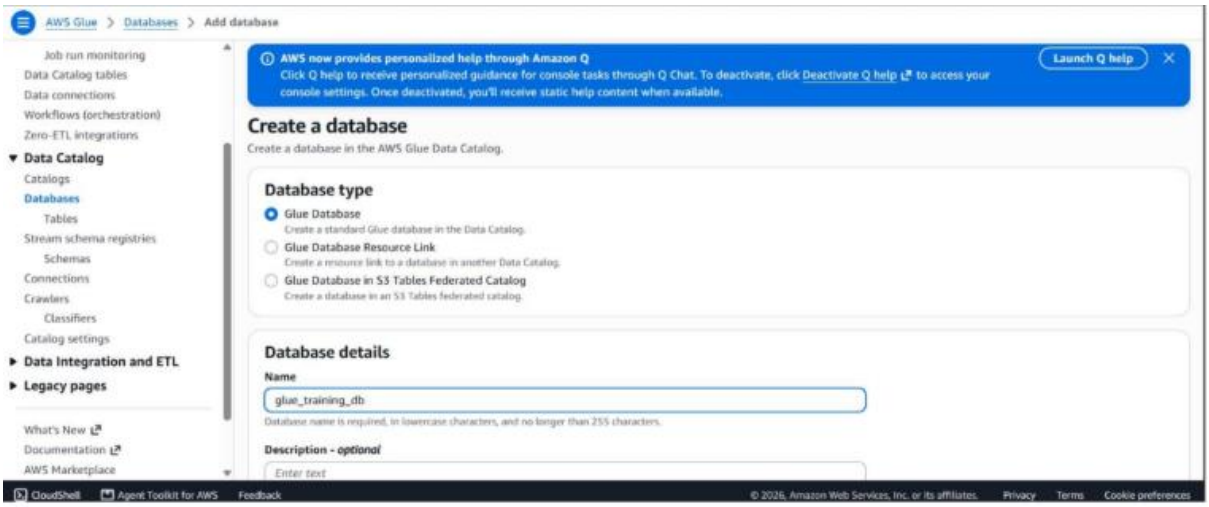
# 1.AWS Glue Console



# 2.Data Catalog

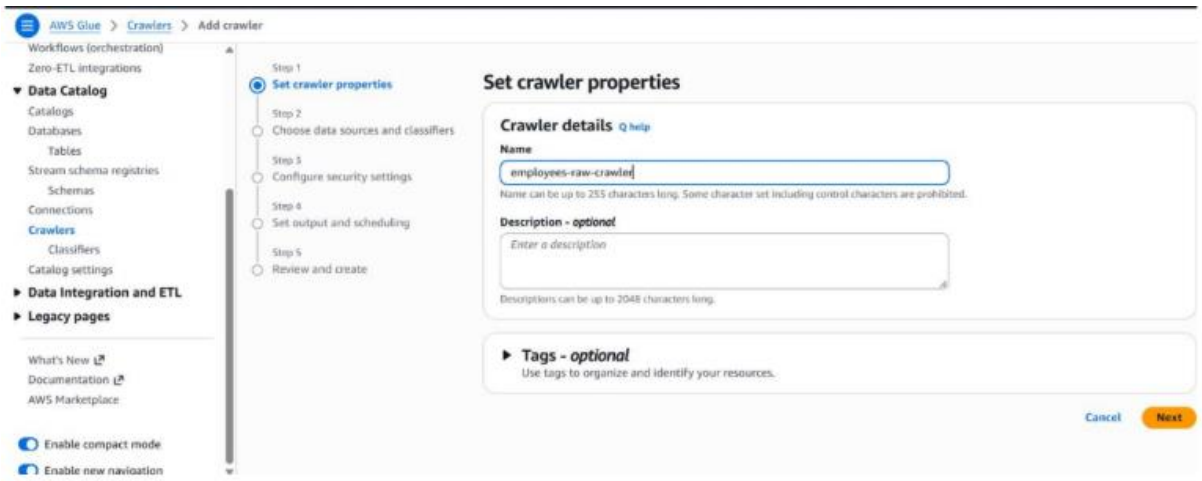


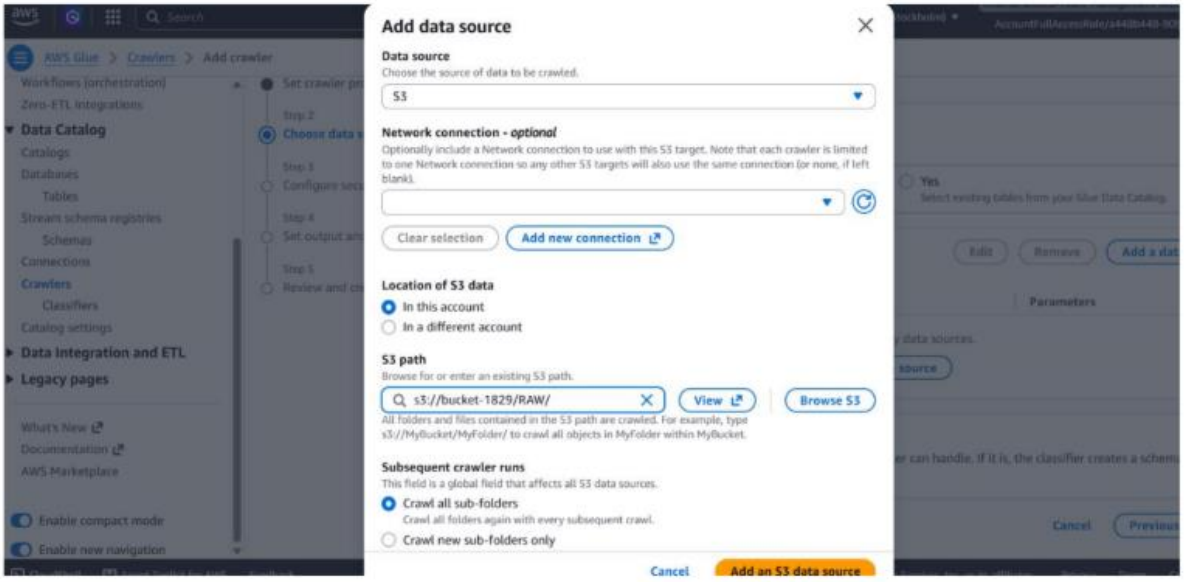

# 3.Database created



# Glue Crawler

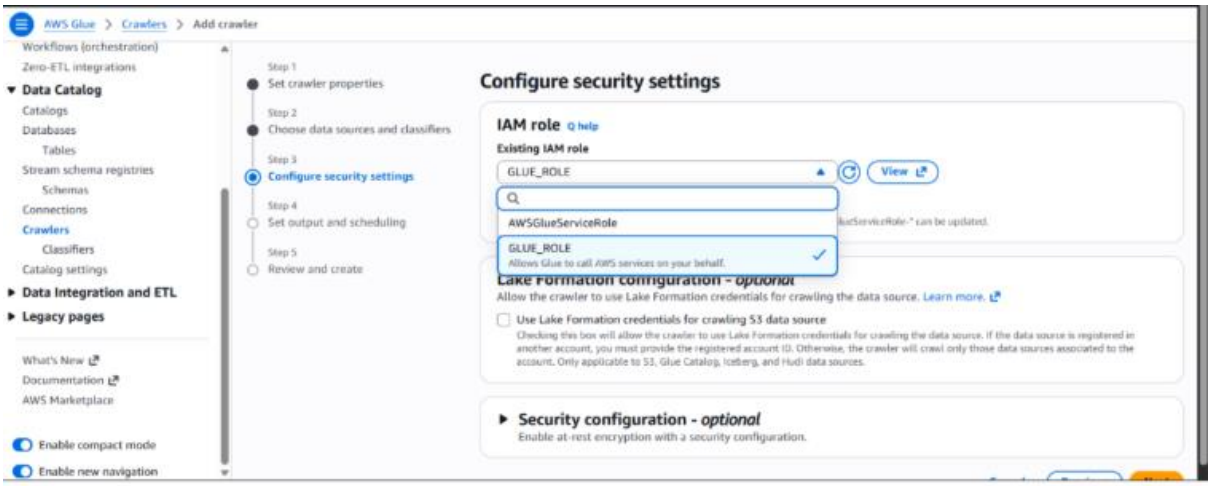
## 1. Crawler name



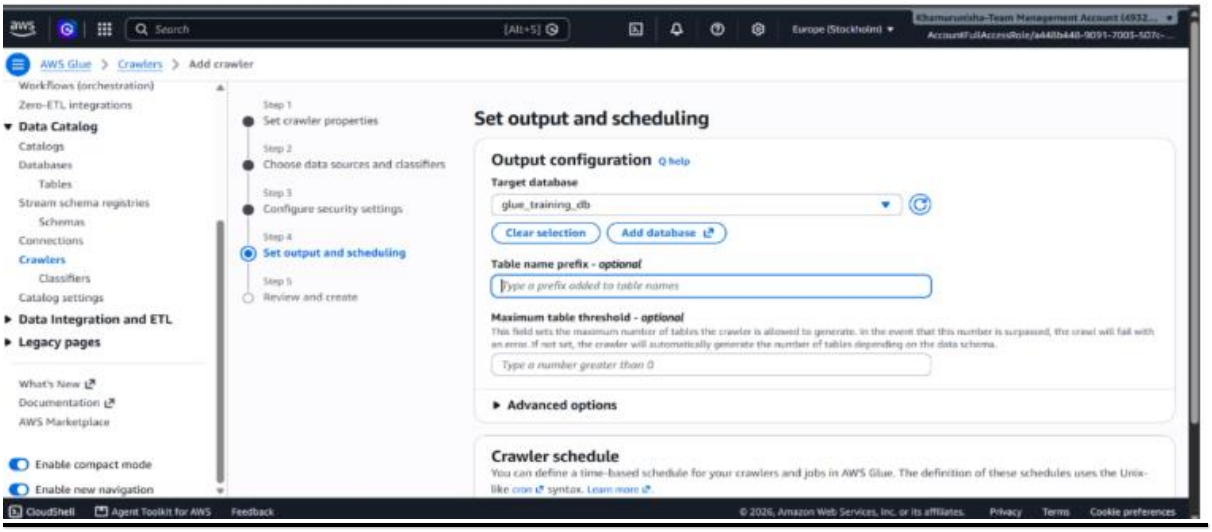




role

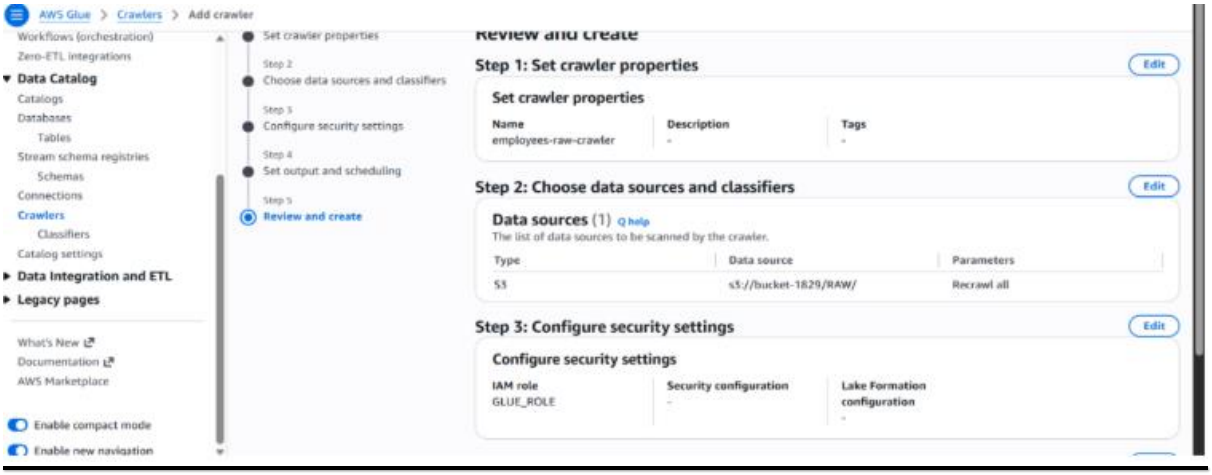
# IAM role




# Target database




# Review and create

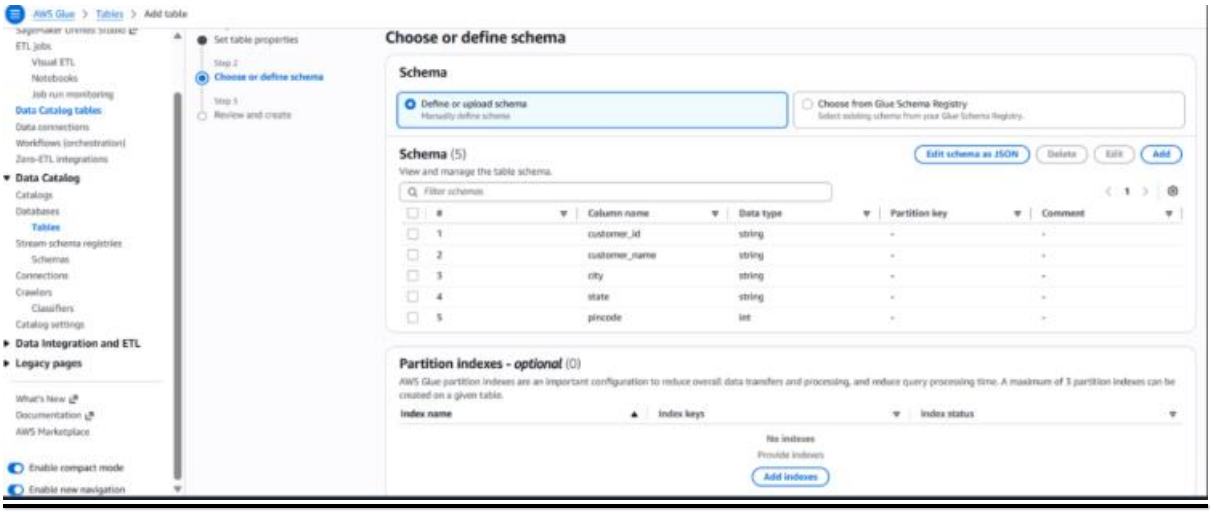


# Glue Data Catalog

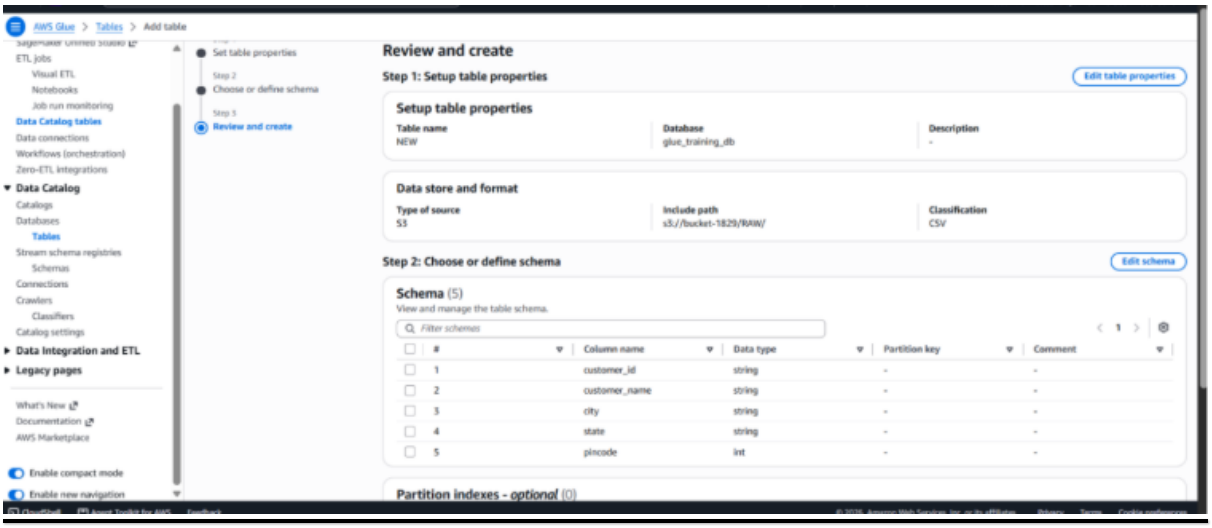
# Create the table manually




# Finish the schema

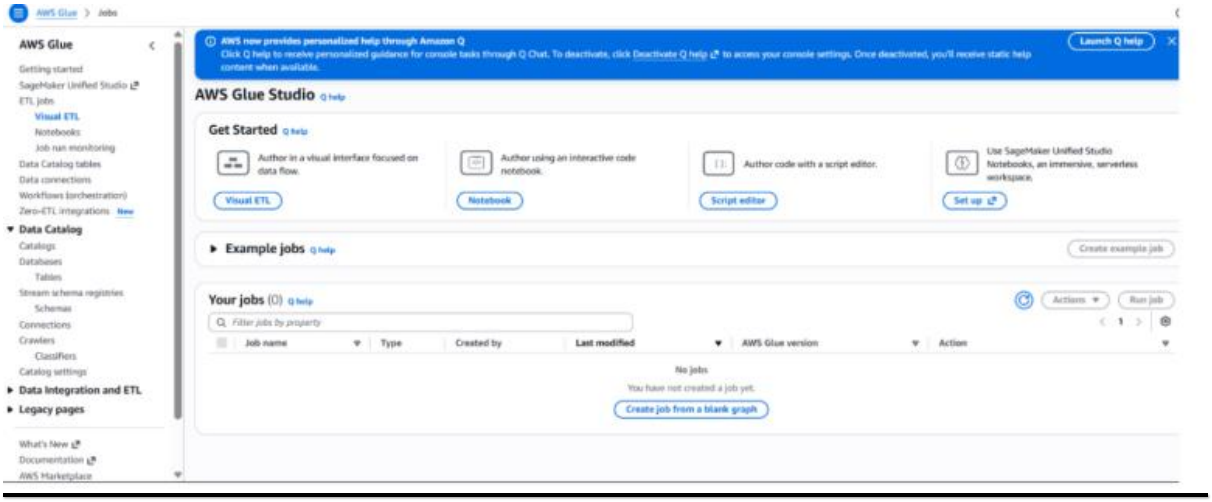


# Review and create

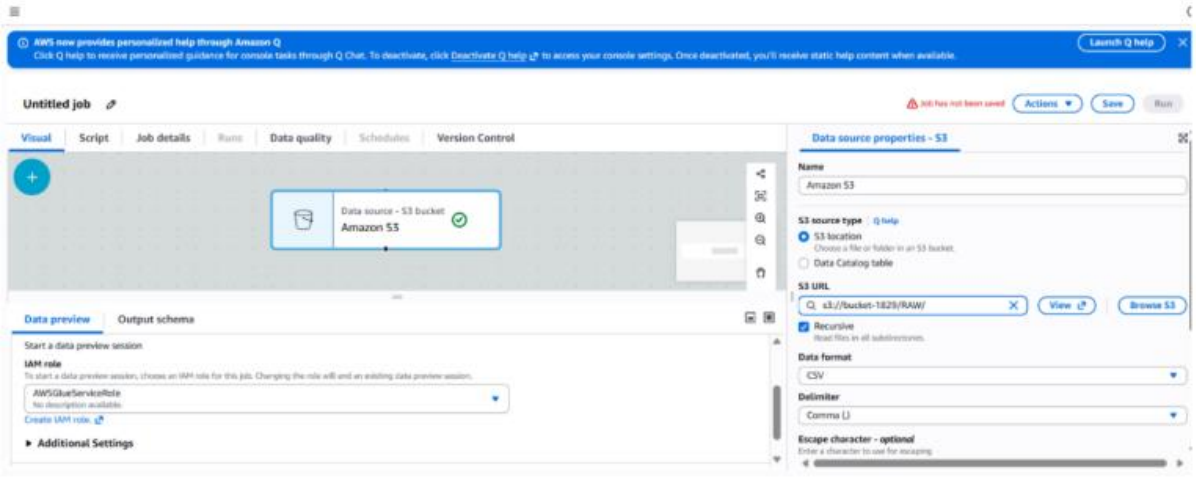



# Glue Visual ETL

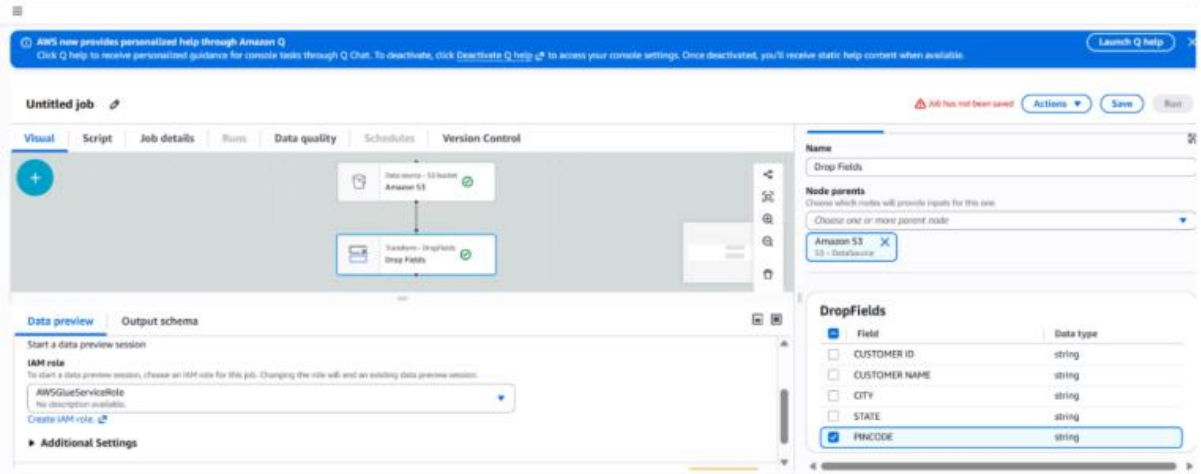
# The complete workflow:



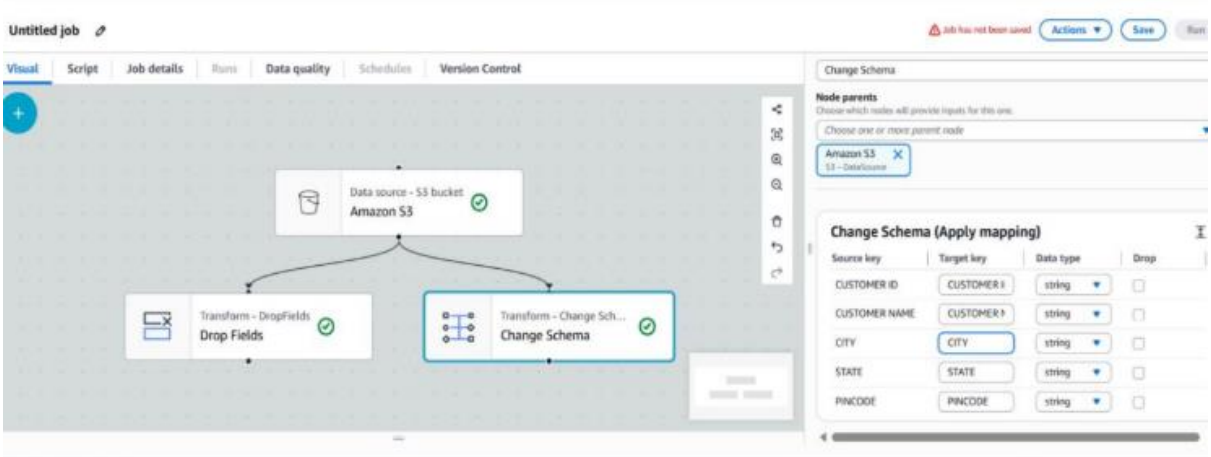
## Create Source



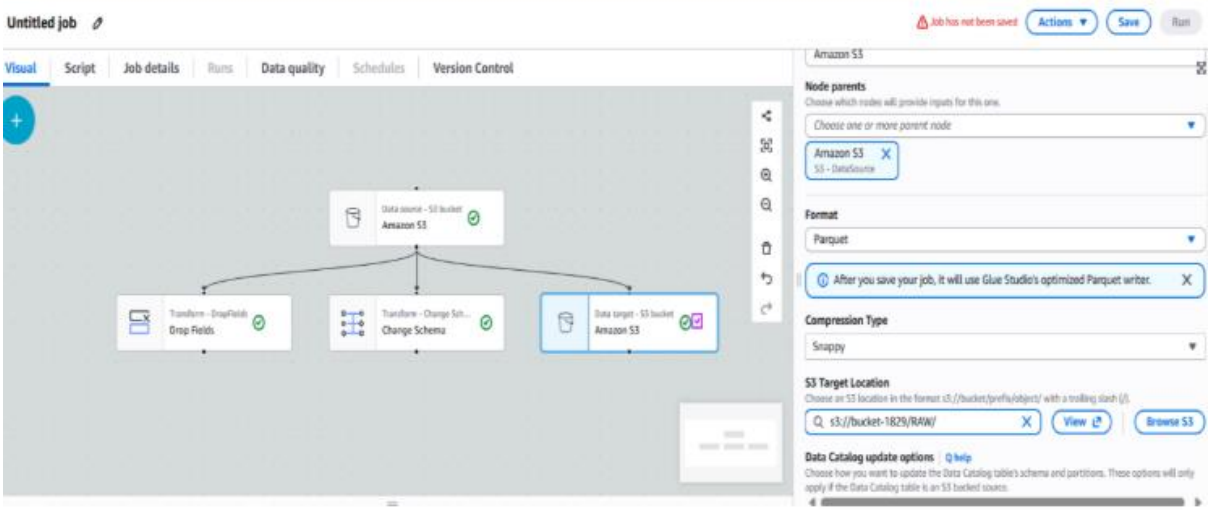

## Glue Transformation(Drop Fields)



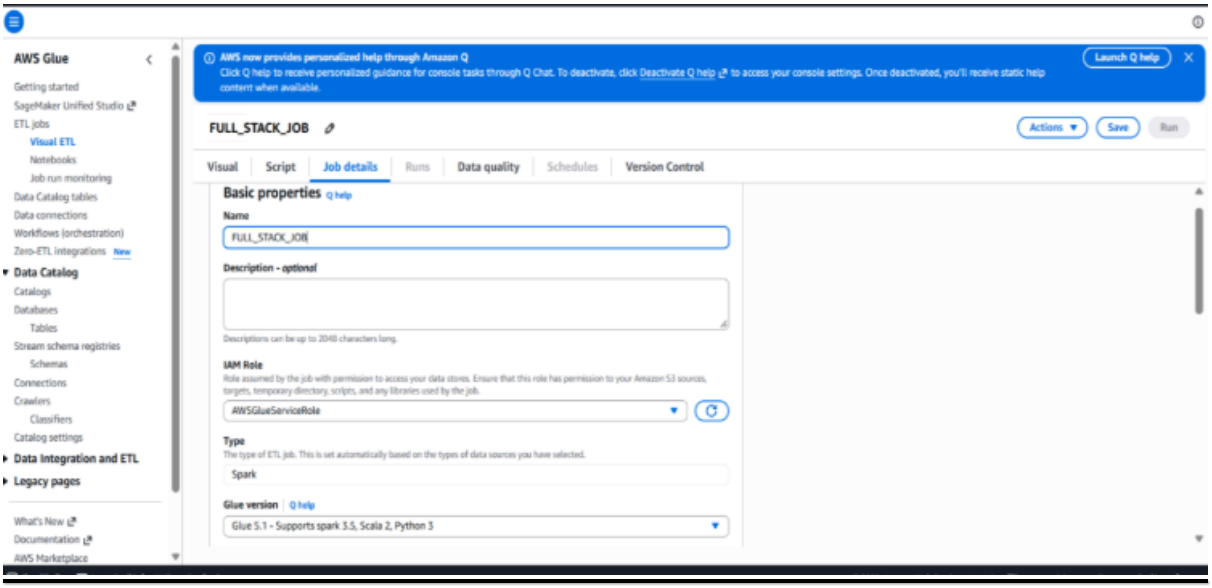

## Glue Transformation



# S3 Target Configuration




# Glue ETL Job Execution

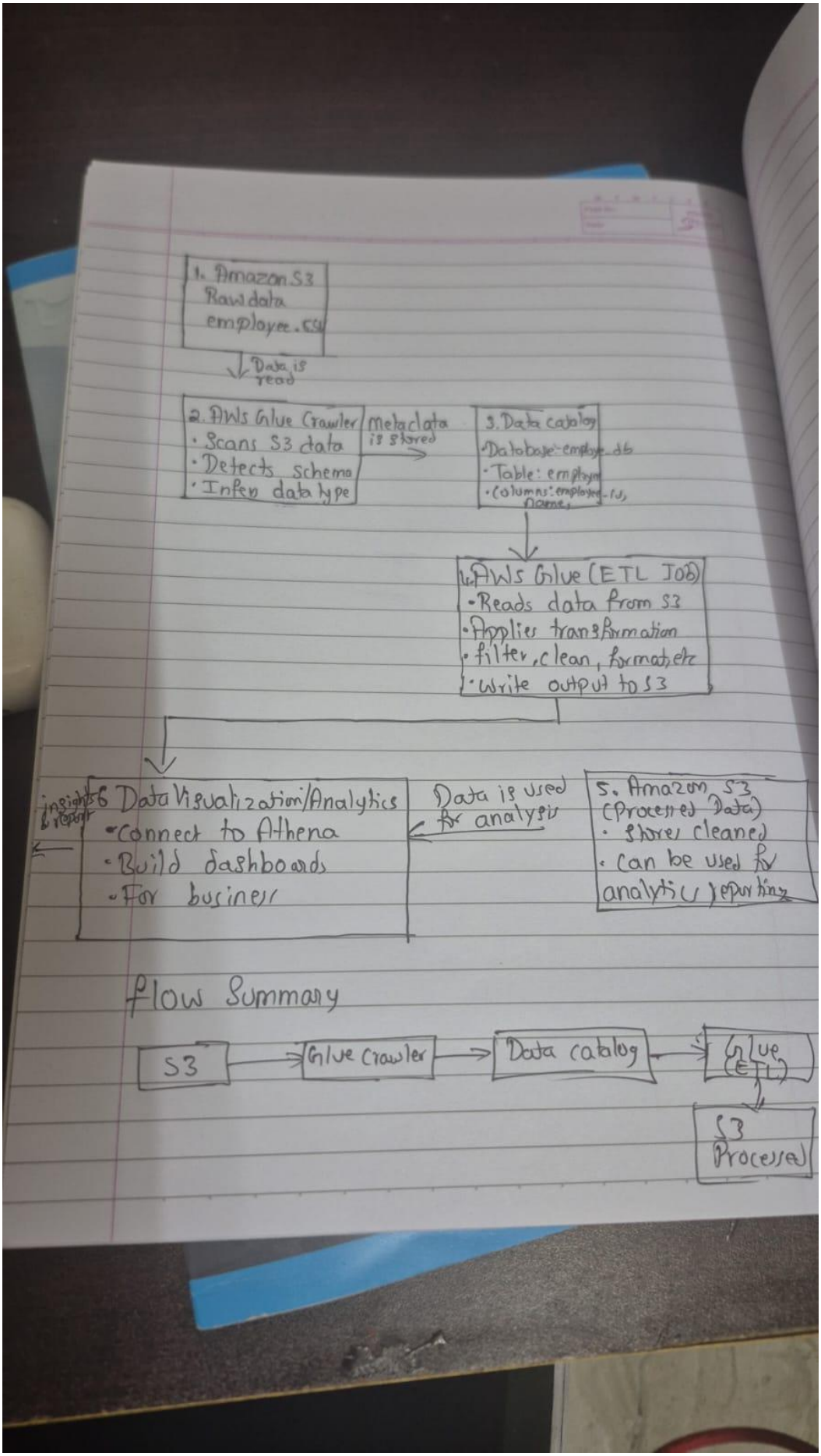


# Final Explanation

- Amazon S3 (Raw Data): First, we upload the customer CSV file to an Amazon S3 bucket. It contains customer ID, name, city, state, and PIN code details.
- AWS Glue Crawler: The crawler scans the uploaded CSV file and automatically identifies the column names, data types, and structure of the data.
- AWS Glue Data Catalog: It stores the metadata, including table names, column details, data types, and the location of the raw data. This allows AWS Glue to access and process the data easily.
- AWS Glue Visual ETL: In this step, we create an ETL job using the visual interface. We connect the source data, transformation steps, and target S3 location.
- Data Transformation: We process the raw data according to our requirements. For example, we can remove unwanted columns and retain only the necessary customer information.
- Output Format (Parquet): After transformation, the data is converted into Apache Parquet format. Parquet is a columnar storage format that supports efficient storage and analytical queries.
- Amazon S3 (Curated Data): The transformed data is stored in the curated/ folder inside the S3 bucket in Parquet format.
- Final Result: The raw customer CSV data is converted into clean, structured, and optimized data that is ready for further analysis, reporting, and business intelligence.

  End-to-End Data Flow

  Customer CSV File → Amazon S3 (Raw Data) → AWS Glue Crawler → Glue Data Catalog → AWS Glue Visual ETL → Data Transformation → Convert to Parquet Format → Amazon S3 (Curated Data) → Ready for Analysis and Reporting.


1. Amazon S3
Raw data
employee.csv
Data is read
2. AWS Glue Crawler
• Scans S3 data
• Detects schema
• Infer data type
Metadata is stored
3. Data catalog
• Database: employee_db
• Table: employee
• Columns: employee_id, name,
4. AWS Glue (ETL JOB)
• Reads data from S3
• Applies transformation
• filter, clean, format, etc
• Write output to S3
6. Data Visualization/Analytics
• Connect to Athena
• Build dashboards
• For business
insights
Data is used for analysis
5. Amazon S3 (Processed Data)
• Stores cleaned
• Can be used for analytics / reporting
Flow Summary
S3
Glue Crawler
Data catalog
Glue (ETL)
S3 Processed